Fully Blind Hyperspectral Unmixing Without Explicit Mixing Models or Endmember Initialization

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***Abstract*—Autoencoder-based methods have gained prominence in blind hyperspectral unmixing (BHU), with most approaches based on predefined mixing assumptions, such as the linear mixing model (LLM). However, real-world scenes often involve complex nonlinear light-material interactions that are difficult to model explicitly, leading to systematic biases in endmember and abundance estimation when these assumptions deviate from the actual mixing mechanism. In this article, we extend conventional autoencoder-based BHU by jointly and adaptively learning the mixing mechanism directly from hyperspectral images (HSIs) while estimating endmembers and abundances, and propose a novel BHU framework termed unmixing Transformer (UNTR). In particular, an adaptive mixing mechanism modeling module is designed to parameterize the mixing mechanism, in which a specially designed transformer fuses heterogeneous endmember and abundance information. Furthermore, we develop an endmember query mechanism integrated into the adaptive mixing mechanism modeling module under a distribution alignment constraint, enabling robust endmember estimation under random initialization. As the first attempt to adaptively model the mixing mechanism as data-driven network parameters, UNTR achieves remarkable accuracy in both abundance and endmember estimation. Without relying on predefined mixing assumptions and well-initialized endmembers, UNTR exhibits strong adaptability to complex real-world scenarios. Our code will be available at https://github.com/Preston-Dong/UNTR**

***Index Terms*—Adaptive mixing mechanism modeling, blind hyperspectral unmixing (BHU), endmember queries, Transformer.**

I. Introduction

LIMITED spatial resolution and the infinitely complex and variable relationships among ground objects result in mixed pixels in hyperspectral images (HSIs), concealing both pure materials and their distribution [1], [2], [3]. Blind hyperspectral unmixing (BHU) addresses this issue by simultaneously decomposing the mixed pixels into a series of pure materials (endmembers) and their corresponding proportions (abundances) [4]. Currently, BHU has been increasingly applied to other critical technologies, such as HSI change detection [5], [6], and super-resolution [7], [8], playing a key role in mineral detection, environmental monitoring, and land-use analysis. The state-of-the-art BHU methods are based on autoencoders [3], which map the input HSI to a latent abundance space through an encoder and then reconstruct it to the original data space using predefined mixing assumptions. Depending on these mixing assumptions, autoencoder-based BHU can be broadly categorized into linear and nonlinear unmixing approaches.

The linear mixing model (LMM) [9] is a simple and widely used assumption. When applied in autoencoders, it represents endmembers as fully connected layer weights and reconstructs the input HSI by multiplying these weights with the estimated abundances. The neural network autoencoder DAEU for BHU, proposed in [10], uses a single spectrum as input and LMM for reconstruction, laying the groundwork for modern approaches. In [11], a deep autoencoder network (DAEN), consisting of a stacked autoencoder and a variational autoencoder, is introduced. An entropic descent archetypal analysis algorithm is proposed in [12]. Nevertheless, the aforementioned algorithms are pixel-level methods [13], [14], lacking spatial–spectral information. In contrast, image-level BHU has gained increasing popularity due to its ability to leverage abundant spatial–spectral information. To incorporate the spatial information of HSIs, a spectral–spatial mixing model called CNNAEU based on a convolutional neural network (CNN) autoencoder is proposed in [15]. DeepTrans, proposed in [16], introduces a Transformer for BHU, constructing a nonlocal feature-dependent encoding structure that uses CNN features as input to the Transformer. To fully leverage hyperspectral bands and spatial information, a multiview spatial–spectral two-stream network MSSSNet is proposed in [17]. In [18], the SWC-Net combines a stationary wavelet transform with CNN for BHU. Considering pixel-level spatial continuity, the ViT-based SRViT is proposed in [19]. Despite effectively utilizing spatial–spectral information [20], [21], [22], [23], these methods are built on LMM, which may not be suitable for complex and nonlinear scenes.

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Nonlinear mixing models (NMMs) provide a more realistic representation of the mixing process, as light-material interactions are inherently complex [24]. By accounting for second-order scatterings, bilinear mixing models (BMMs) [25], [26], [27] constitute an important subclass of NMMs, including the Fan model [28], the generalized bilinear model (GBM) [29], [30], and the Polynomial post-NMM (PPNMM) [31]. In addition, the Hapke model [32] is commonly adopted for intimate mixtures, while the multilinear mixing model (MLM) [33] provides a more generalized formulation that incorporates higher order scatterings. Beyond explicit physical models, manifold-based [34], and kernel-based [35] approaches have also been explored as model-free alternatives. Recent advancements in BHU have introduced the above NMMs into autoencoders to further enhance unmixing performance [36], [37], [38], [39], [40], [41], [42]. In [36], a post-nonlinear approach is introduced into the decoder structure, where the product of estimated endmembers and abundances serves as input to subsequent fully connected layers for spectrum refinement. The research in [37] employs a 3-D convolutional network to extract local spatial–spectral features, which are then fed into a fully connected network-based autoencoder, enhancing the decoding structure with a nonlinear network. The HapkeCNN model [38] integrates the Hapke model with an autoencoder, while [39] proposes a graph attention convolutional autoencoder (GACAE) that leverages both long-range and short-range spatial information, building its decoder based on PPNMM. In addition, Fang et al. [40] incorporates MLM into autoencoders, developing pixel-level and patch-level models to estimate abundances, endmembers, and transition probabilities simultaneously. In [41], an extended multilinear mixing (EMLM) model-inspired dual-stream network is proposed. This approach employs the alternating direction method of multipliers (ADMMs) for encoder construction and uses the EMLM assumption [43] for the decoder. Similarly, a deep autoencoder-based augmented network (DAAN) [42] is proposed based on MLM. By incorporating the diffusion model (DM) and EMLM into the network, a diffusion-enhanced autoencoder (DEAE) [44] is designed.

However, existing autoencoder-based methods inherently rely on predefined mixing assumptions [16], [20], [21], [22], [38], [40], [41], typically tailored to specific scenarios, such as the Fan model for bilinear interactions. In practice, real-world scenarios often exhibit intricate and unpredictable light-material interactions that are difficult to represent explicitly [1], [2]. When the assumed mixing model deviates from the actual mixing mechanism, systematic biases in endmember and abundance estimation become unavoidable.

Moreover, by treating the weights of fully connected layers as endmembers, most autoencoder-based methods, regardless of whether they adopt linear or nonlinear mixing assumptions, largely depend on well-initialized endmembers. Many approaches [10], [40], [45] adopt distinct learning rates for the encoder and decoder, typically assigning a lower learning rate to the endmembers. As a result, the endmembers are updated only marginally and tend to remain close to their initial values. Furthermore, in the absence of a robust endmember initialization strategy, such as vertex component analysis (VCA) [46], the performance of these methods degrades significantly and exhibits substantial uncertainty, due to the limited constraints imposed on endmember estimation when the weights of fully connected layers are treated as endmembers. To address this issue, recent autoencoder-based approaches have progressively decoupled endmembers from fully connected layer parameters, thereby enabling endmember optimization from random initialization during training under conventional constraints [23], [47], [48], [49], [50]. In particular, UNMamba [23] and MiSiCNet [47] employ the minimum volume constraint to facilitate endmember convergence. In [48] and [49], endmember estimation is reformulated as the product of the reconstructed image and the transpose of the abundance matrix. Nevertheless, this formulation inherently relies on the LLM, which limits its capability to capture the more complex mixing mechanisms commonly observed in real-world data. Besides, in [50], the total variation (TV) loss is introduced to enforce spectral smoothness. However, when predefined mixing assumptions substantially deviate from the actual mixing mechanism, endmember optimization from random initialization can still be adversely affected. The core challenge lies in robustly aligning randomly initialized endmembers with the image data distribution by effectively capturing the correlations between endmembers and the observed hyperspectral data.

In summary, BHU can be substantially improved by developing a framework that adaptively models the mixing mechanism from HSIs and robustly maps randomly initialized sequences to endmembers in accordance with the image data distribution. Nevertheless, satisfying these requirements remains a major challenge for existing autoencoder-based approaches. Although Transformer-based models have been widely adopted in BHU to extract effective spatial–spectral features and map them to abundances [16], [45], [51], [52], their potential extends beyond feature extraction. In particular, its cross-attention mechanism provides a powerful means of fusing heterogeneous sources of information [53], making it especially suitable for integrating endmember and abundance representations. Motivated by this insight, an adaptive mixing mechanism modeling approach that leverages the Transformer's attention mechanism is proposed to improve the accuracy of endmember and abundance estimation. The main contributions of this article are as follows.

1) We extend conventional autoencoder-based BHU by jointly and adaptively learning the mixing mechanism directly from HSIs while estimating endmembers and abundances, and propose a novel BHU framework termed unmixing Transformer (UNTR) that achieves accurate endmember and abundance estimation without relying on predefined mixing assumptions or well-initialized endmembers.
2) Distinct from existing BHU approaches that primarily emphasize spatial–spectral feature extraction, this article designs an adaptive mixing mechanism modeling module, in which a specially designed transformer fuses heterogeneous endmember and abundance information,

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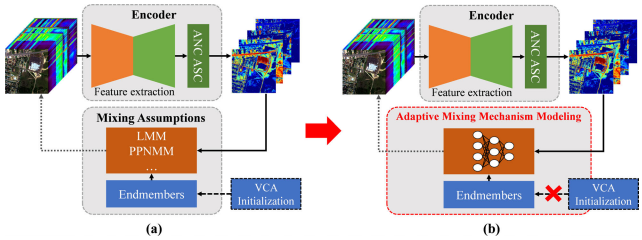


Fig. 1. Comparison between the typical autoencoder-based approach and the proposed adaptive mixing mechanism modeling. (a) Typical autoencoder-based approach. (b) Proposed approach.

making the first attempt to parameterize the mixing mechanism adaptively through network parameters.

3) By analyzing the reliance on well-initialized endmembers and performance degradation caused by random initialization, we propose an endmember query mechanism integrated into the adaptive mixing mechanism modeling module under a distribution alignment constraint to enable accurate endmember estimation from random initialization.

The remainder of this article is organized as follows. Section II introduces the proposed methodology, covering motivation, network structure, and the use of constraints. Section III presents a detailed overview of the simulation data, real data, parameter settings, and experimental results from various comparison methods. Section IV provides a discussion on the validity of UNTR. Finally, Section V offers a summary and conclusion.

II. Proposed Method

This section begins with an overview of the motivation and a general description of the UNTR methodology. It then provides a detailed explanation of the UNTR network structure, the implementation of the adaptive mixing mechanism modeling, and the endmember query mechanism with the distribution alignment constraint. Finally, the loss function and the auxiliary constraints that ensure the physical meaning of the endmembers and abundances are described.

A. Motivation and Overview

The typical autoencoder-based BHU approach, shown in Fig. 1(a), extracts spatial–spectral features from the input HSI and maps them to abundances using the abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC). Afterward, the input HSI is reconstructed by predefined mixing assumptions, such as LMM or PPNMM, to enable self-supervised learning [3]. Despite the emergence of various mixing assumptions [24], [28], [29], [33], the actual mixing mechanism in real-world scenarios is complex and difficult to describe intuitively. When the assumed mixing assumption deviates significantly from the actual mixing mechanism, it inevitably leads to biases in endmember and abundance estimation. Let $\mathbf{I} \in \mathbb{R}^{L\times H\times W}$ represent the HSI to be unmixed, with spatial dimensions H and W, and L spectral bands. In addition, let $\mathbf{E} \in \mathbb{R}^{R\times L}$ represent the endmember matrix, R the number of endmembers, and $\mathbf{M} \in \mathbb{R}^{R\times H\times W}$ the corresponding abundances. The generalized nonlinear mixing mechanism in HSI can be expressed as follows:

$$\mathbf{I} = \mathbf{\Phi}(\mathbf{E}, \mathbf{M}) + \mathbf{N}, \quad \text{s.t.} \quad \mathbf{M} \geq 0 \text{and } \mathbf{1}_R^{\mathrm{T}}\mathbf{M} = \mathbf{J} \tag{1}$$

where $\mathbf{N} \in \mathbb{R}^{L\times H\times W}$ denotes the additive noise and $\mathbf{J}$ is a matrix consisting entirely of ones. Considering the physical meaning of abundance, $\mathbf{M} \geq 0$and $\mathbf{1}_R^{\mathrm{T}}\mathbf{M} = \mathbf{J}$ represent ANC and ASC, respectively [24]. Besides, $\mathbf{\Phi}$ is the mixing mechanism, a function acting on the endmembers $\mathbf{E}$ and parameterized by the corresponding abundances $\mathbf{M}$ [3]. Unlike current methods, leveraging the fitting capability of deep learning to learn $\mathbf{\Phi}$, as shown in Fig. 1(b), enables adaptive mixing mechanism modeling, effectively mitigating biases in endmember and abundance estimation caused by discrepancies between predefined mixing assumptions and the actual mixing mechanism.

Moreover, because most autoencoder-based methods treat the weights of fully connected layers as endmembers, it is difficult to impose explicit constraints directly on endmember estimation. As a consequence, well-initialized endmembers, such as those obtained by VCA, are often required to accelerate convergence and avoid poor local optima. From the perspective of traditional optimization methods, endmember constraints naturally facilitate convergence and enable reliable endmember estimation even under random initialization. By decoupling endmember estimation from fully connected layers and incorporating conventional constraint strategies into the autoencoder framework, endmember estimation can be more effectively guided during training, thereby fully leveraging the representation learning capability and optimization flexibility of self-supervised deep learning.

Therefore, UNTR is proposed by adaptively modeling the mixing mechanism under randomly initialized endmembers. UNTR does not rely on predefined mixing assumptions or well-initialized endmembers, demonstrating exceptional adaptability and estimation accuracy across various scenarios. In particular, UNTR comprises an encoder for abundance estimation, an adaptive mixing mechanism modeling module, and an endmember query mechanism module, as shown in Fig. 2.

B. Encoder for Abundance Estimation

Similar to existing autoencoder-based approaches, the encoder of UNTR is designed to estimate abundances by extracting rich spatial–spectral features and mapping them to a latent abundance space [3]. A wide range of feature extraction techniques, from CNNs [15], [18] and Transformers [16], [19] to the recently introduced Mamba [23], have demonstrated strong representational capabilities. Nevertheless, the primary contribution of this work lies in adaptive mixing mechanism modeling. Within this framework, feature extraction serves a supportive role, providing sufficient representation without becoming the performance bottleneck. Therefore, we adopt a simplified U-Net with multiscale fusion as the UNTR encoder.

The specific structure is shown in Fig. 3. It consists of a simplified U-Net network with two downsampling and two upsampling layers to extract and fuse multiscale spatial–spectral features. Since the Softmax activation function

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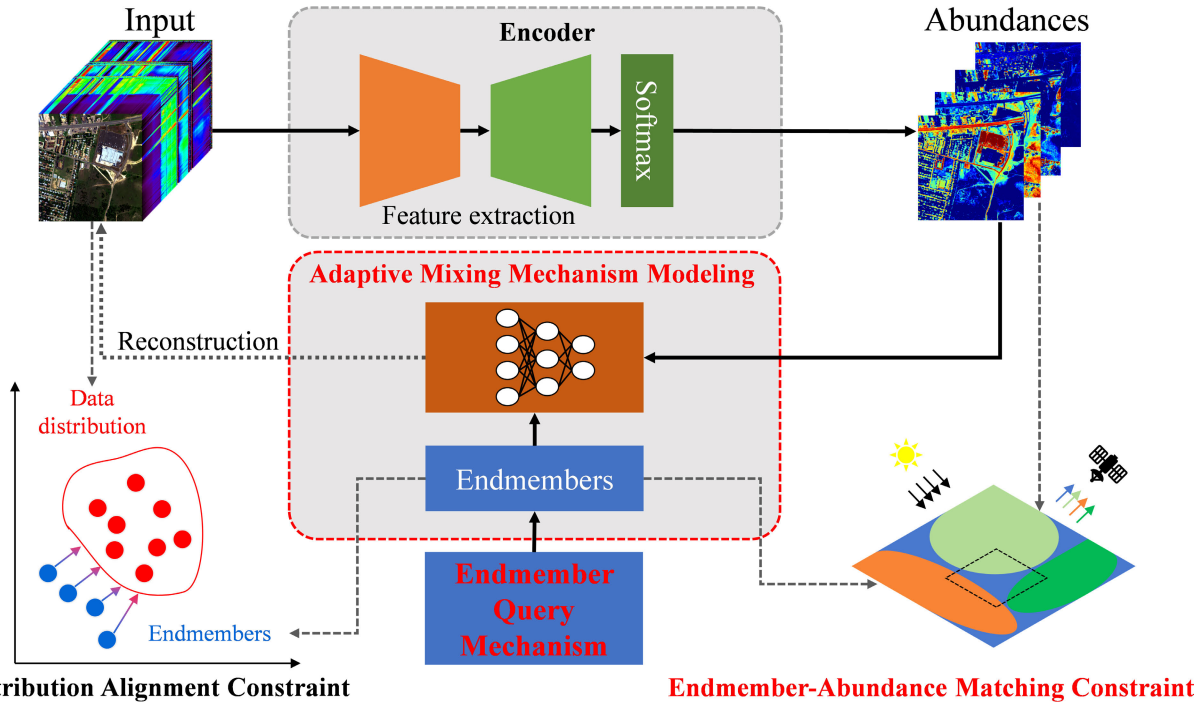


Fig. 2. Overview of the proposed UNTR. It comprises an encoder for abundance estimation, an adaptive mixing mechanism modeling module, and an endmember query mechanism module. Besides, a distribution alignment constraint is designed to guide the convergence of randomly initialized endmembers toward the image distribution, while an endmember-abundance matching constraint makes the abundances correspond to endmembers.

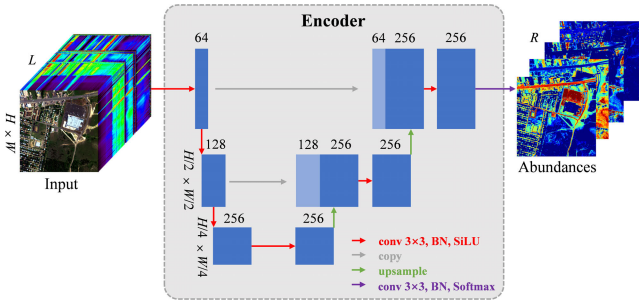


Fig. 3. Network structure of the encoder. It adopts a simplified U-Net network to extract and fuse multiscale spatial–spectral features.

is suitable for ANC and ASC, the final abundances are generated from the spatial–spectral features after applying it

$$\hat{\mathbf{M}}_{(i,h,w)} = \text{Softmax}\left(\omega \mathbf{X}_{(i,h,w)}\right) = \frac{e^{\omega \mathbf{X}_{(i,h,w)}}}{\sum_j e^{\omega \mathbf{X}_{(j,h,w)}}} \tag{2}$$

where $\mathbf{X}_{(i,h,w)}$ denotes the ith feature to be activated by Softmax at spatial location (h, w). Considering abundance sparsity, the scale factor ω is used to control scaling [15] and is set to 3.5 in our experiments.

C. Endmember Query Mechanism

In existing autoencoder frameworks, the reliance on well-initialized endmembers, as well as the perturbations and uncertainties induced by random initialization, fundamentally stems from the nonconvex nature of the BHU optimization problem, which is exacerbated by the limited constraints imposed on the weights of fully connected layers. On the one hand, random initialization often drives the optimization toward locally optimal rather than globally optimal endmember estimates, while representing each endmember with a single sequence further increases estimation variance. On the other hand, randomly initialized endmembers are poorly aligned with the image data distribution, thereby increasing training difficulty. As a result, random initialization introduces considerable instability and a strong dependence on favorable initialization, such as VCA.

Intuitively, optimization should aim to reduce the variance of endmember estimation to approach globally optimal solutions, while simultaneously ensuring alignment between the endmember and the image distribution. Accordingly, we propose an endmember query mechanism that represents each endmember as multiple trainable random sequences combined through a specific function $\mathbf{F}(\cdot)$. This design reduces estimation variance and facilitates the learning of robust estimated endmembers that are less sensitive to initialization, thereby mitigating the perturbations caused by relying on a single random sequence. Furthermore, it aligns the endmembers, represented by these multiple random sequences, with the image data distribution during training, allowing for their optimization toward the direction of the HSI. To simplify the design of the endmember query mechanism, UNTR employs a straightforward average function as $\mathbf{F}$

$$\hat{\mathbf{E}}_i = \frac{1}{K} \sum_{j=i\cdot K}^{(i+1)\cdot K} \mathbf{Q}_{\text{E}}^{(j)}, \quad i = 0, 1, \ldots, R-1 \tag{3}$$

where $\mathbf{Q}_{\text{E}} \in \mathbb{R}^{R\cdot K\times L}$ denote endmember queries, which are trainable random sequences, $\hat{\mathbf{E}}_i$ represents the ith endmember, and $K \in \mathbb{Z}^+$ is a scale factor controlling the

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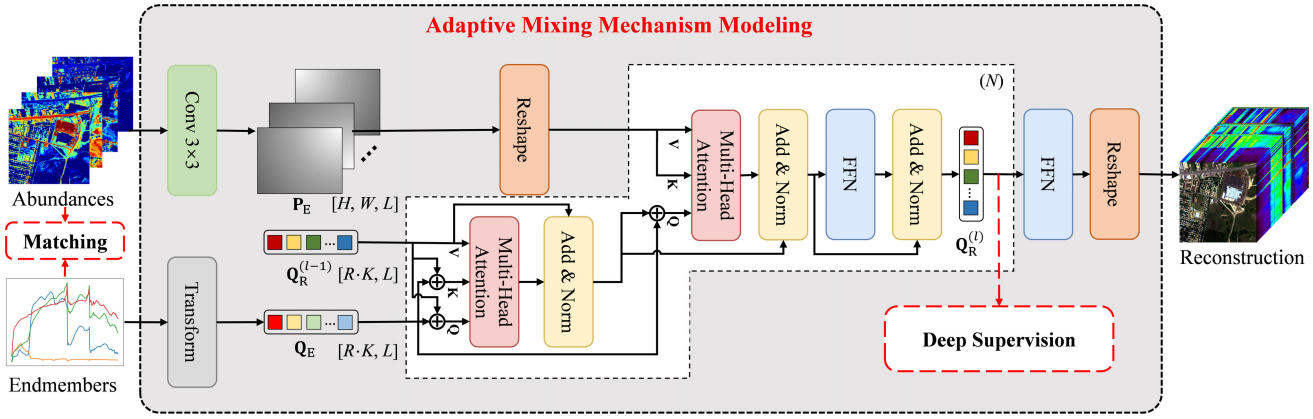


Fig. 4. Network structure of the adaptive mixing mechanism modeling module. It comprises N identically structured networks, where the network parameters represent the mixing mechanism, fusing the estimated endmembers and abundances to reconstruct the HSI.

endmember queries for each endmember. Notably, when $K = 1$, the endmember queries correspond directly to the estimated endmembers. Representing endmembers by averaging multiple random sequences effectively reduces fluctuations in endmember estimation. Moreover, using multiple random sequences allows endmembers to be characterized from diverse perspectives, improving the model adaptability to the global data distribution and facilitating the learning of varied endmembers.

Meanwhile, to facilitate the learning of randomly initialized endmembers and better align them with the image data distribution during training, we introduce the minimum volume constraint as a distribution alignment constraint [47], in which the mean spectrum is used to represent the HSI distribution

$$\mathcal{L}_{\mathrm{DA}} = \frac{1}{R}\sum_{i=1}^{R}\left\|\hat{\mathbf{E}}_i - \bar{\mathbf{I}}\right\|_2^2 \tag{4}$$

where $\bar{\mathbf{I}}$ denotes the mean spectrum and $\|\cdot\|_2$ represents the l_2 norm. This constraint guides each endmember toward the hyperspectral data space, reducing reliance on well-initialized endmembers. Notably, formulating the endmember convergence problem as a data distribution alignment task allows additional strategies that are beneficial to distribution alignment to be naturally incorporated. Further distribution alignment constraints are discussed in Section IV-J.

D. Adaptive Mixing Mechanism Modeling

The adaptive mixing mechanism modeling module, central to UNTR, reconstructs HSI by combining endmembers with their corresponding abundances, using neural network parameters to represent the inherent mixing mechanism in HSI. This approach transforms the mixing mechanism into the learning of decoder parameters by designing effective decoding structures that integrate endmembers and abundances. Inspired by DETR [54], where queries effectively describe images, such as object queries in object detection, this article proposes reconstruction queries $\mathbf{Q}_R \in \mathbb{R}^{R\cdot K\times L}$. They are designed to inscribe the image and are integrated into BHU, leveraging the Transformer's attention mechanism to iteratively optimize them for reconstructing the input HSI. The proposed adaptive mixing mechanism modeling module, as shown in Fig. 4, comprises N identically structured networks, progressively optimizing the reconstruction queries through each stage. Initially, it maps the estimated abundances to per-pixel embeddings $\mathbf{P}_E \in \mathbb{R}^{H\times W\times L}$, which provide abundance information to the network structure

$$\mathbf{P}_E = \mathrm{SiLU}\left(\mathrm{Conv}\left(\mathbf{M}\right)\right). \tag{5}$$

In parallel, the estimated endmembers are transformed to produce queries containing endmember information $\mathbf{Q}_{EM} \in \mathbb{R}^{R\cdot K\times L}$

$$\mathbf{Q}_{EM} = \mathrm{Transform}\left(\mathbf{E}\right) \tag{6}$$

where this Transform can be implemented in various ways, such as a fully connected layer or a convolution. Considering that the multiple random sequences $\mathbf{Q}_E$ in the endmember query mechanism can capture endmember properties, it is straightforward to use $\mathbf{Q}_E$ to represent $\mathbf{Q}_{EM}$. This also implies that the Transform here can be viewed as the inverse of $\mathbf{F}$ in (3), serving as one of many possible transforms. Assuming that $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ represent the Query, Key, and Value vectors in the Transformer attention mechanism, respectively, the output $\mathbf{Z}$ can be expressed as

$$\mathbf{Z} = \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{d}}\right)\mathbf{V} \tag{7}$$

where $\sqrt{d}$ is used to control the scaling scale. Considering the absence of positional embeddings in the Transformer structure itself, reconstruction queries in the mixing mechanism modeling module lack mutual positional relationships between bands. Since endmember queries effectively describe ground objects, they are well-suited to serve as positional embeddings for reconstruction queries. Therefore, the proposed Transformer decoder incorporates endmember queries as the position embeddings, first constructing the self-attention mechanism for the reconstruction queries and then optimizing the reconstruction queries by fusing per-pixel embeddings.

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One of the Transformer decoders in the mixing mechanism learning module can be represented as

$$\begin{cases} \mathbf{Z}^{(l,1)} = \text{Softmax}\left[\dfrac{\left(\mathbf{Q}_{\text{E}} + \mathbf{Q}_{\text{R}}^{(l-1)}\right)\left(\mathbf{Q}_{\text{E}} + \mathbf{Q}_{\text{R}}^{(l-1)}\right)^{\text{T}}}{\sqrt{d}}\right]\mathbf{Q}_{\text{R}}^{(l-1)} \\ \mathbf{T}^{(l,1)} = \mathbf{Q}_{\text{R}}^{(l-1)} + \text{LN}\left(\mathbf{Z}^{(l,1)}\right) \\ \mathbf{Z}^{(l,2)} = \text{Softmax}\left[\dfrac{\left(\mathbf{Q}_{\text{E}} + \mathbf{T}^{(l,1)}\right)\mathbf{P}_{\text{E}}^{\text{T}}}{\sqrt{d}}\right]\mathbf{P}_{\text{E}} \\ \mathbf{T}^{(l,2)} = \mathbf{T}^{(l,1)} + \text{LN}\left(\mathbf{Z}^{(l,2)}\right) \end{cases} \tag{8}$$

where $\mathbf{Z}^{(l,1)}$ and $\mathbf{Z}^{(l,2)}$ denote the self-attention and cross-attention mechanisms in the lth decoder, respectively. In practice, the multihead attention (MHA) mechanism is employed instead. $\mathbf{T}^{(l,\cdot)}$ represents the summation and layer normalization (LN) operations. When $l = 1$, $\mathbf{Q}_{\text{R}}^{(l-1)}$ is initialized with all elements set to zero. Then, the updated $\mathbf{Q}_{\text{R}}^{(l)}$ can be expressed as

$$\mathbf{Q}_{\text{R}}^{(l)} = \mathbf{T}^{(l,2)} + \text{LN}\left[\text{FFN}\left(\mathbf{T}^{(l,2)}\right)\right]. \tag{9}$$

In which FFN means a feed-forward network. After that, the final reconstruction queries $\mathbf{Q}_{\text{R}}^{(N)}$ are processed through FFN and reshaped to generate the reconstructed HSI

$$\hat{\mathbf{I}} = \text{Reshape}\left[\text{FFN}\left(\mathbf{Q}_{\text{R}}^{(N)}\right)\right] \tag{10}$$

where $\hat{\mathbf{I}}$ represents the reconstructed HSI. In our experiments, N is set to 2, indicating the use of two identically structured networks.

E. Loss Function

BHU does not rely on groundtruth for model training, making the construction of the loss function critical. A common BHU loss function is the reconstruction loss combined with the abundance sparsity constraint [55], [56], [57], [58]. Considering that the spectral angle distance (SAD) captures scale invariance [10] and the mean square error (mse) captures numerical error, both are employed in this article to construct the reconstruction loss $\mathcal{L}_{\text{rec}}$

$$\begin{aligned} \mathcal{L}_{\text{rec}} &= \mathcal{L}_{\text{SAD}} + \alpha\mathcal{L}_{\text{MSE}} \\ &= \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\arccos\left[\frac{\mathbf{I}_{(h,w)}^{T}\hat{\mathbf{I}}_{(h,w)}}{\left\|\mathbf{I}_{(h,w)}\right\|_2\left\|\hat{\mathbf{I}}_{(h,w)}\right\|_2}\right] \\ &\quad + \frac{\alpha}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\left\|\mathbf{I}_{(h,w)} - \hat{\mathbf{I}}_{(h,w)}\right\|_2^2 \end{aligned} \tag{11}$$

where (h, w) denotes the spatial location and $\|\cdot\|_2$ represents the l_2 norm. α is the weight of mse, which is set to 1 in our experiments. For the abundance sparsity constraint, we use $l_{1/2}$ regularization to enforce this sparsity

$$\mathcal{L}_{\text{sparse}} = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\left\|\hat{\mathbf{M}}_{(h,w)}\right\|_{1/2}. \tag{12}$$

When combined with the proposed distribution alignment constraint (4) to enable endmember learning under random initialization, the constructed loss function becomes

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \beta\mathcal{L}_{\text{sparse}} + \gamma\mathcal{L}_{\text{DA}} \tag{13}$$

in which β and γ are hyperparameters. In the experiments, β and γ are set to 0.0001 and 1, respectively.

F. Endmember-Abundance Matching Constraint

The proposed UNTR, while capable of adaptively modeling the mixing mechanism from HSI, lacks direct alignment between the estimated endmembers and abundances. To address this, the endmember-abundance matching constraint based on LMM is proposed, as illustrated in Fig. 2. LMM can serve as a rough estimate of the endmembers and abundances, with more refined estimation typically provided by complex NMMs [9]. This characteristic makes LMM well-suited for endmember and abundance matching, where the focus is on achieving a reliable match. The LMM-based reconstructed image can be represented as

$$\hat{\mathbf{I}}_{\text{EAM}} = \hat{\mathbf{E}}^{\text{T}}\hat{\mathbf{M}}. \tag{14}$$

Then, the endmember-abundance matching constraint $\mathcal{L}_{\text{EAM}}$ can be expressed as

$$\mathcal{L}_{\text{EAM}} = \mathcal{L}_{\text{rec}}\left(\mathbf{I}, \hat{\mathbf{I}}_{\text{EAM}}\right). \tag{15}$$

In addition, this constraint implies that UNTR adopts an LMM-guided mixing mechanism modeling strategy, which is consistent with existing NMMs that incorporate a linear term. This design enables nonlinear interactions among endmembers while simultaneously preserving the direct representation of each material's spectral signature.

G. Deep Supervision

The reconstruction queries in UNTR are progressively optimized by N identical structures, and the output $\mathbf{Q}_{\text{R}}^{(l)}$ of the lth structure also contributes to HSI recovery. To aid in better optimization of the reconstruction queries, this article proposes a reconstruction query auxiliary loss based on deep supervision [59]

$$\mathcal{L}_{\text{AUX}} = \frac{1}{N}\sum_{l=1}^{N-1}\mathcal{L}_{\text{rec}}\left(\mathbf{I}, \hat{\mathbf{I}}^{(l)}\right) \tag{16}$$

where $\hat{\mathbf{I}}^{(l)}$ denotes the reconstructed HSI from $\mathbf{Q}_{\text{R}}^{(l)}$ through the shared operation in (10). By incorporating deep supervision, additional supervision of the intermediate structures in the adaptive mixing mechanism modeling module helps optimize the reconstruction queries more effectively, improving the model's robustness. It also prevents the overfitting of subsequent structures, reducing the network's reliance on the final loss function and accelerating the model convergence.

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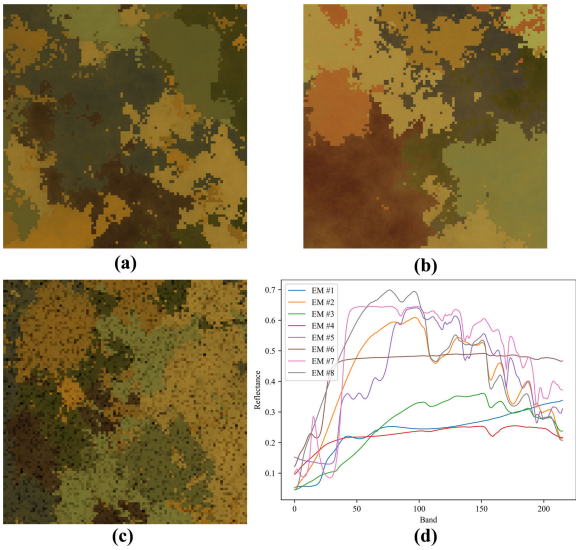


Fig. 5. False-color images of the synthetic datasets and the endmembers used. (a) LMM-based synthetic dataset. (b) Fan model-based synthetic dataset. (c) MLM-based synthetic dataset. (d) Endmembers.

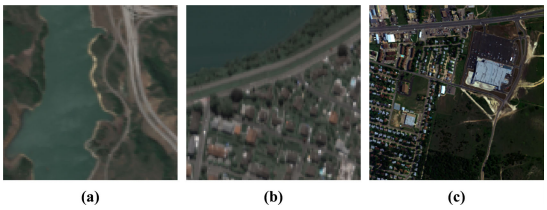


Fig. 6. True-color images of three real datasets used in the experiments. (a) Jasper Ridge. (b) Apex. (c) Urban.

III. Experimental Results

A. Data Description

Experiments were conducted on three synthetic datasets and three real datasets. For the synthetic datasets, as shown in Fig. 5, the spatial dimensions in the experiments are set to 100×100, and the abundances are generated using the HSI synthesis toolbox (HYDRA) with Gaussian fields. The endmembers are sourced from the USGS Digital Spectral Library, comprising 216 bands. To simulate complex real-world scenarios, eight endmembers are employed to synthesize three datasets with different mixing assumptions, LMM, Fan model, and MLM, under a signal-to-noise ratio of 40 dB. In particular, for the MLM simulation, the pixelwise transition probability generation strategy described in [33] and [40] is employed.

The true-color images of these three real datasets are shown in Fig.6. The Jasper Ridge dataset [60] comprises 100×100 pixels with 224 bands covering the range from 380 to 2500 nm. After removing affected bands, 198 bands are retained for BHU validation, with four endmembers: road, soil, water, and tree. Apex [61] is a relatively uncommon unmixing dataset, recently utilized in [16]. This dataset consists of 110×110 pixels with 285 bands ranging from 413 to 2420 nm. It contains four labeled endmembers: water, tree, road, and roof. Furthermore, the Urban dataset [58] represents a large-scale scene consisting of 307×307 pixels, making it well-suited for effectively validating unmixing algorithms. Each pixel originally contains 210 bands covering the range from 400 to 2500 nm, with 162 bands retained for unmixing verification. This dataset provides multiple reference versions; in this work, the four-endmember reference is adopted for validation.

B. Experimental Setup

In comparing algorithms, this article reproduces a wide range of conventional and deep learning methods. Notable conventional methods include FCLSU [28], SCLSU [62], and NUSAL [63]. The deep learning methods are categorized into pixel-level unmixing and image-level unmixing. For pixel-level methods, DAEU [10] with SAD as the loss, SIDAEU [14] with spectral information divergence (SID) as the loss, MLAEM [36] focusing on PPNMM, and MLM-1DAE [40] based on the MLM assumption are selected. For image-level unmixing, we compare with recent methods, including Transformer-based approaches such as DeepTrans [16] and SRViT [19], the stationary wavelet transform-based SWCNet, and the newly proposed A2SN and A2SAN [48] without endmember initialization. Except for UNTR, A2SN, and A2SAN, all other methods use the VCA [46] algorithm to initialize endmembers. All deep learning methods are implemented using the PyTorch framework, with model training conducted on an RTX 4090 graphics card. Given that the UNTR method proposed in this article lacks endmember initialization, the number of epochs is set to 1000 for model convergence. The initial learning rate is set to 0.003, decreasing to 90% of its value every 45 epochs.

To quantitatively evaluate the performance of BHU methods, the conventional root mse (RMSE) is used as the evaluation metric for abundances, whereas the SAD is employed for endmembers. Besides, the average RMSE (aRMSE) and average SAD (aSAD) are used as overall evaluation metrics, as detailed below

$$\text{aRMSE} = \frac{1}{R}\sum_{i=1}^{R}\sqrt{\frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\left(\mathbf{M}_{(i,h,w)}-\hat{\mathbf{M}}_{(i,h,w)}\right)^2} \tag{17}$$

$$\text{aSAD} = \frac{1}{R}\sum_{i=1}^{R}\arccos\left(\frac{\mathbf{E}_i^{\mathrm{T}}\hat{\mathbf{E}}_i}{\|\mathbf{E}_i\|_2\left\|\hat{\mathbf{E}}_i\right\|_2}\right) \tag{18}$$

where $\mathbf{E}_i$ denotes the ith endmember, and $\|\cdot\|_2$ represents the l_2 norm.

C. Synthetic Data Results

The experimental results for the three synthetic datasets are presented in Table I, where the endmember errors along with the corresponding abundance errors for each category are provided in detail, and the average value is used as the overall evaluation metric.[1] The bold font highlights the best results obtained from the experiments. In addition, the overall evaluation metrics, aRMSE and aSAD, are marked with circular numbers to visually indicate both the optimal and suboptimal experimental results.

[1]Visualization results of the proposed UNTR on these three synthetic datasets are provided in Appendix A.

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TABLE I

RMSE AND SAD RESULTS FOR THE THREE SYNTHETIC DATASETS. MINIMUM VALUES FOR EACH ENDMEMBER ARE IN BOLD, WITH THE AVERAGE AS THE OVERALL METRIC. THE BEST RESULT IS SHOWN IN RED WITH ①, AND THE SECOND-BEST IN BLUE WITH ②

Datasets	Metrics	Methods	EM #1	EM #2	EM #3	EM #4	EM #5	EM #6	EM #7	EM #8	Average
Linear mixing model	RMSE	DAEU [10]	0.1032	0.1266	0.1091	0.1430	0.0674	0.2014	0.0805	0.0609	0.1115
		SIDAEU [14]	0.0926	0.2147	0.1186	0.2319	0.1101	0.1625	0.0507	0.1283	0.1387
		MLAEM [36]	0.2153	0.1791	0.2242	0.2828	0.2185	0.1009	0.2241	0.1284	0.1967
		DeepTrans [16]	0.1473	0.0787	0.1010	0.1487	0.0509	0.0341	0.0526	0.0756	0.0861
		MLM-1DAE [40]	0.0712	0.1857	0.0322	0.1747	0.0685	0.0470	**0.0338**	0.0755	0.0861
		A2SN [48]	0.1431	0.1785	0.2573	0.1745	0.2006	0.1684	0.1010	0.1233	0.1683
		A2SAN [48]	0.0854	0.1749	0.1183	0.1301	0.0922	0.1606	0.2175	0.1620	0.1426
		SWCNet [18]	0.0365	0.1210	0.0406	0.0756	0.0766	0.0480	0.0516	0.0756	0.0657
		SRViT [19]	0.0558	0.0755	0.0459	0.0426	0.0530	0.0464	0.0352	**0.0524**	0.0508②
		UNTR	**0.0283**	**0.0406**	**0.0289**	**0.0218**	**0.0337**	**0.0340**	0.0411	0.0584	**0.0358**①
	SAD	DAEU [10]	0.0771	0.0365	0.0674	0.0756	0.0335	0.0368	0.0231	0.0205	0.0463
		SIDAEU [14]	0.0806	0.0674	0.0657	0.0808	0.0376	0.0551	0.0238	0.0453	0.0571
		MLAEM [36]	0.1591	0.0939	0.2178	0.1853	0.1236	0.1693	0.0652	0.1326	0.1434
		DeepTrans [16]	0.0556	**0.0218**	0.0525	0.0574	0.0175	**0.0145**	0.0170	0.0265	0.0328
		MLM-1DAE [40]	0.0718	0.1241	**0.0204**	0.0565	0.0228	0.0373	**0.0088**	**0.0082**	0.0437
		A2SN [48]	0.0576	0.1176	0.1378	**0.0198**	0.1048	0.0684	0.0794	0.0572	0.0804
		A2SAN [48]	0.0593	0.0758	0.0278	0.0251	**0.0112**	0.0722	0.1520	0.0144	0.0547
		SWCNet [18]	0.0224	0.1137	0.0138	0.0206	0.0154	0.0598	0.0140	0.0197	0.0349
		SRViT [19]	**0.0193**	0.0312	0.0289	0.0246	0.0174	0.0200	0.0115	0.0158	**0.0211**①
		UNTR	0.0504	0.0367	0.0389	0.0314	0.0176	0.0260	0.0161	0.0156	0.0291②
Fan model	RMSE	DAEU [10]	0.1423	0.1076	0.2427	0.2112	0.1202	0.1982	0.1003	0.0620	0.1481
		SIDAEU [14]	0.1077	0.2185	0.1994	0.1539	0.1184	0.1669	0.0685	0.1198	0.1441
		MLAEM [36]	0.1949	0.2154	0.2600	0.2403	0.1449	0.1517	0.1312	0.1498	0.1860
		DeepTrans [16]	0.0464	0.0678	0.0750	0.0764	0.0483	0.0585	0.0256	0.0582	0.0570
		MLM-1DAE [40]	0.0970	0.1732	0.1428	0.1121	0.0365	0.0793	0.0763	0.1295	0.1058
		A2SN [48]	0.0981	0.2029	0.2019	0.1677	0.2127	0.2380	0.1228	0.1713	0.1769
		A2SAN [48]	0.2332	0.1276	0.1444	0.1377	0.1146	0.1756	0.1314	0.0622	0.1408
		SWCNet [18]	0.0532	0.0685	0.0409	0.0300	0.0611	0.0851	0.0564	0.0619	0.0571
		SRViT [19]	0.0400	0.0448	0.0322	0.0372	**0.0275**	0.0328	0.0180	**0.0300**	0.0328②
		UNTR	**0.0277**	**0.0295**	**0.0260**	**0.0297**	0.0425	**0.0196**	**0.0176**	0.0390	**0.0289**①
	SAD	DAEU [10]	0.1492	0.0292	0.1540	0.0923	0.0560	0.0870	0.0267	0.0180	0.0765
		SIDAEU [14]	0.0934	0.0664	0.1414	0.0843	0.0580	0.0547	0.0317	0.0586	0.0736
		MLAEM [36]	0.1783	0.1016	0.2106	0.2169	0.1456	0.2189	0.0664	0.1868	0.1656
		DeepTrans [16]	0.0552	0.0115	0.0502	0.0585	0.0198	0.0266	0.0095	0.0273	0.0323
		MLM-1DAE [40]	0.1360	0.0654	0.1706	0.0884	0.0142	0.0468	0.0308	0.0998	0.0815
		A2SN [48]	0.0437	0.1265	0.0891	0.0404	0.1387	0.0859	0.1445	0.0720	0.0926
		A2SAN [48]	0.1385	0.0237	0.0632	0.0406	0.0138	0.0320	0.1393	0.0148	0.0582
		SWCNet [18]	**0.0198**	0.0140	0.0156	0.0256	**0.0063**	**0.0096**	**0.0038**	**0.0116**	0.0133②
		SRViT [19]	0.0252	**0.0069**	**0.0094**	**0.0219**	0.0077	0.0133	0.0073	0.0122	**0.0130**①
		UNTR	0.0569	0.0127	0.0287	0.0415	0.0231	0.0178	0.0145	0.0175	0.0266
Multilinear mixing model	RMSE	DAEU [10]	0.1249	0.1938	0.1786	0.1799	0.1163	0.1939	0.1640	0.2362	0.1734
		SIDAEU [14]	0.1403	0.1683	0.1919	0.1915	0.1693	0.1742	0.1230	0.1524	0.1639
		MLAEM [36]	0.2371	0.2633	0.2024	0.1924	0.2344	0.2444	0.1343	0.2565	0.2206
		DeepTrans [16]	0.1022	0.2250	0.0965	0.0776	0.2138	0.3016	0.1592	0.1421	0.1647
		MLM-1DAE [40]	0.1616	0.2370	0.1698	0.1715	0.1856	0.2098	0.1459	0.2068	0.1860
		A2SN [48]	0.2307	0.3105	0.2288	0.2133	0.1633	0.2160	0.1313	0.2143	0.2135
		A2SAN [48]	0.0614	0.1182	0.0471	0.1572	0.0495	0.1517	0.1344	0.0871	0.1008②
		SWCNet [18]	0.1015	0.2768	0.1487	0.1151	0.1164	0.2255	0.1292	0.1658	0.1599
		SRViT [19]	0.1442	0.2957	0.2077	0.1296	0.1441	0.1346	0.1577	0.2870	0.1876
		UNTR	**0.0564**	**0.0837**	**0.0403**	**0.0417**	**0.0386**	**0.0816**	**0.0325**	**0.0468**	**0.0527**①
	SAD	DAEU [10]	0.0615	0.1072	0.1238	0.1277	0.1883	0.0635	0.1756	0.0894	0.1171
		SIDAEU [14]	0.0919	0.1252	0.1891	0.1756	0.1544	0.1489	0.1002	0.0790	0.1331
		MLAEM [36]	0.1089	0.0786	0.1598	0.1897	0.2108	0.1874	0.1679	0.0955	0.1498
		DeepTrans [16]	0.0441	0.1044	0.1749	0.0456	0.1787	0.0482	0.1690	0.0296	0.0993
		MLM-1DAE [40]	0.0653	0.0553	0.0912	0.1715	0.1158	0.2098	0.0921	0.0502	0.0812
		A2SN [48]	0.0483	0.1096	0.1825	0.0995	0.0732	0.0566	0.1447	0.1112	0.1032
		A2SAN [48]	**0.0285**	0.0459	**0.0139**	**0.0317**	0.0467	0.0539	0.1496	0.0447	0.0519②
		SWCNet [18]	0.0366	0.1192	0.1884	0.0928	**0.0121**	0.0271	0.1734	**0.0180**	0.0835
		SRViT [19]	0.1138	0.1009	0.1996	0.1651	0.2287	0.1612	0.2048	0.0289	0.1504
		UNTR	0.0433	**0.0347**	0.0433	0.0488	0.0183	**0.0264**	**0.0196**	0.0203	**0.0318**①

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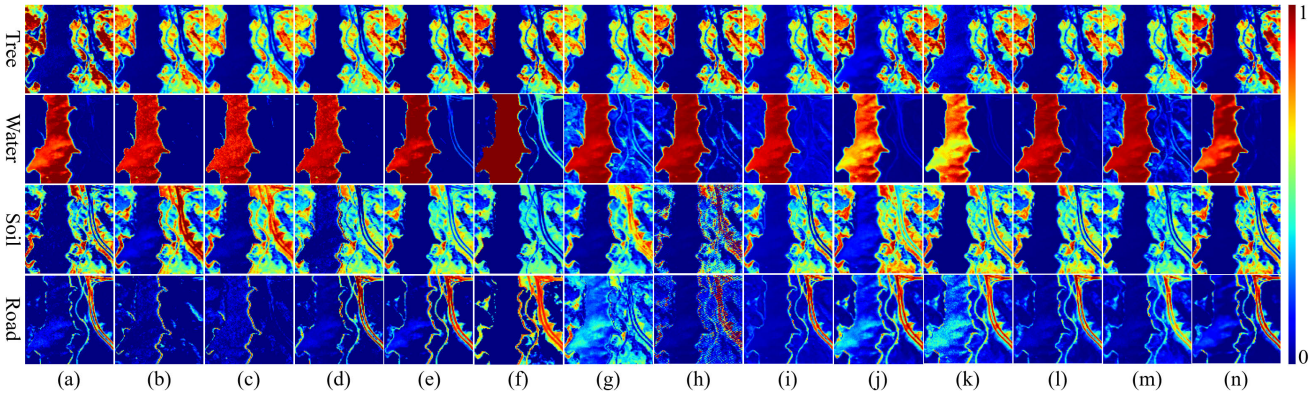


Fig. 7. Estimated abundances of different methods on the Jasper Ridge dataset. (a) Reference. (b) FCLSU. (c) SCLSU. (d) NUSAL. (e) DAEU. (f) SIDAEU. (g) MLAEM. (h) DeepTrans. (i) MLM-1DAE. (j) A2SN. (k) A2SAN. (l) SWCNet. (m) SRViT. (n) UNTR.

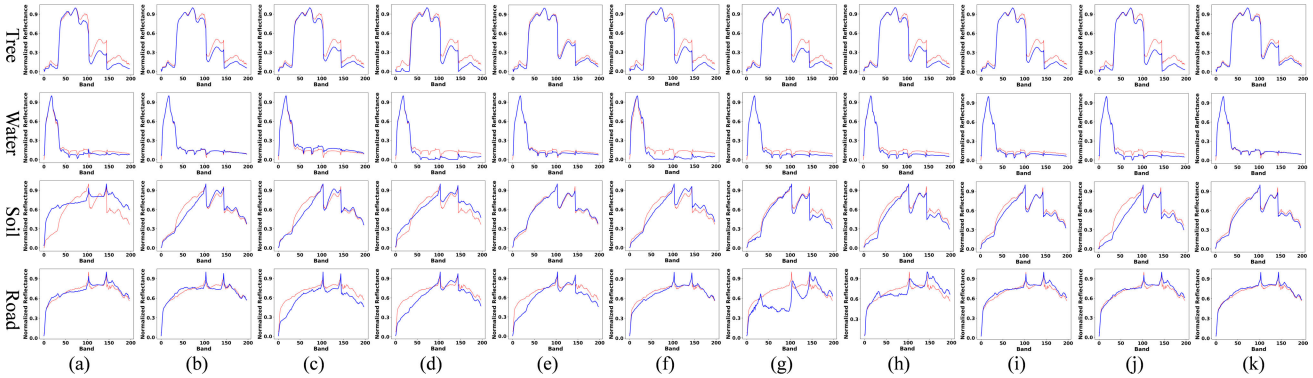


Fig. 8. Estimated endmembers of different methods on the Jasper Ridge dataset. Rows from the first to the last correspond to tree, water, soil, and road, respectively. (a) VCA. (b) DAEU. (c) SIDAEU. (d) MLAEM. (e) DeepTrans. (f) MLM-1DAE. (g) A2SN. (h) A2SAN. (i) SWCNet. (j) SRViT. (k) UNTR.

These synthetic datasets are generated using eight endmembers and exhibit considerable mixing complexity. On the LMM-based synthetic dataset, SRViT achieves the best performance in endmember estimation and the second-best performance in abundance estimation, benefiting from its strong feature extraction capability and a design tailored to the LMM setting with VCA initialization. In contrast, the proposed UNTR attains the best abundance estimation and the second-best endmember estimation using only a simplified U-Net for feature extraction, without relying on well-initialized endmembers provided by VCA. Experimental results for the LMM-based synthetic dataset demonstrate the effectiveness of the proposed mixing mechanism modeling for assumed LMM and validate that the endmember query mechanism can reduce reliance on well-initialized endmembers.

The Fan model-based synthetic dataset, which accounts for second-order scatterings, exhibits a higher degree of nonlinear mixing complexity. On this dataset, UNTR achieves the best abundance estimation and comparable performance in endmember estimation, while SRViT attains the second-best abundance estimation and the best endmember results. Notably, UNTR achieves these state-of-the-art unmixing results without relying on predefined mixing assumptions or well-initialized endmembers.

The MLM-based synthetic dataset is notably more complex, presenting substantial challenges for both endmember and abundance estimation. Nevertheless, the proposed UNTR achieves the best performance with randomly initialized endmembers, outperforming the compared methods. The suboptimal method is A2SAN, attributed to its attention-based spatial–spectral feature extraction capabilities. These results emphasize that the proposed mixing mechanism modeling approach is equally effective for MLM, demonstrating its ability to adapt and accurately capture the mixing mechanism. By providing a more precise representation of the inherent mixing mechanism, it effectively corrects errors in endmember and abundance estimation, thereby improving unmixing accuracy.

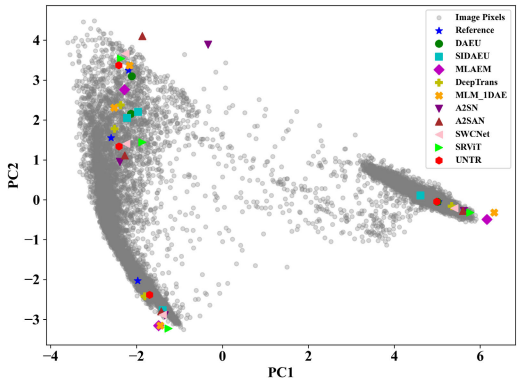


Fig. 9. PCA-based visualization of all pixels, reference endmembers, and estimated endmembers from different methods on the Jasper Ridge dataset.

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TABLE II

RMSE AND SAD RESULTS FOR THE JASPER RIDGE DATASET. MINIMUM VALUES FOR EACH ENDMEMBER ARE IN BOLD, WITH THE AVERAGE AS THE OVERALL METRIC. THE BEST RESULT IS SHOWN IN RED WITH ①, AND THE SECOND-BEST IN BLUE WITH ②

Methods	Initialization	RMSE					SAD				
		Tree	Water	Soil	Road	aRMSE	Tree	Water	Soil	Road	aSAD
FCLSU [28]	VCA	0.1069	0.0786	0.2353	0.2505	0.1678	0.1676	0.1420	0.2705	0.0551	0.1588
SCLSU [62]	VCA	0.1134	0.0767	0.2207	0.2456	0.1641	0.1676	0.1420	0.2705	0.0551	0.1588
NUSAL [63]	VCA	0.1339	0.0776	0.1632	0.0829	0.1144	0.1676	0.1420	0.2705	0.0551	0.1588
DAEU [10]	VCA	0.0825	0.0563	0.1082	0.0808	0.0819②	0.1143	**0.0282**	0.1235	0.0518	0.0794
SIDAEU [14]	VCA	0.0914	0.1450	0.1713	0.1411	0.1372	0.1080	0.1008	0.1159	0.1131	0.1095
MLAEM [36]	VCA	0.1529	0.1651	0.1912	0.2193	0.1821	0.1545	0.2926	0.1356	0.1440	0.1817
DeepTrans [16]	VCA	0.0770	0.0992	0.2069	0.2000	0.1458	0.0562	0.0797	**0.0318**	0.1364	0.0760②
MLM-1DAE [40]	VCA	0.1369	0.0774	0.1310	0.0804	0.1064	0.1531	0.3243	0.1163	0.0508	0.1586
A2SN [48]	–	0.1096	0.1130	0.1254	0.1113	0.1148	0.1287	0.1473	0.0721	0.2568	0.1512
A2SAN [48]	–	0.0940	0.1321	0.1049	0.1276	0.1147	0.1115	0.1447	0.0871	0.1112	0.1136
SWCNet [18]	VCA	0.1109	0.0581	0.1011	0.0632	0.0833	0.1209	0.0955	0.0574	0.0411	0.0787
SRViT [19]	VCA	0.1454	0.1277	0.1427	0.1065	0.1306	0.1697	0.1867	0.1391	0.0435	0.1347
UNTR	Random	**0.0584**	**0.0211**	**0.0811**	**0.0668**	0.0569①	**0.0503**	0.0298	0.0375	**0.0223**	0.0350①

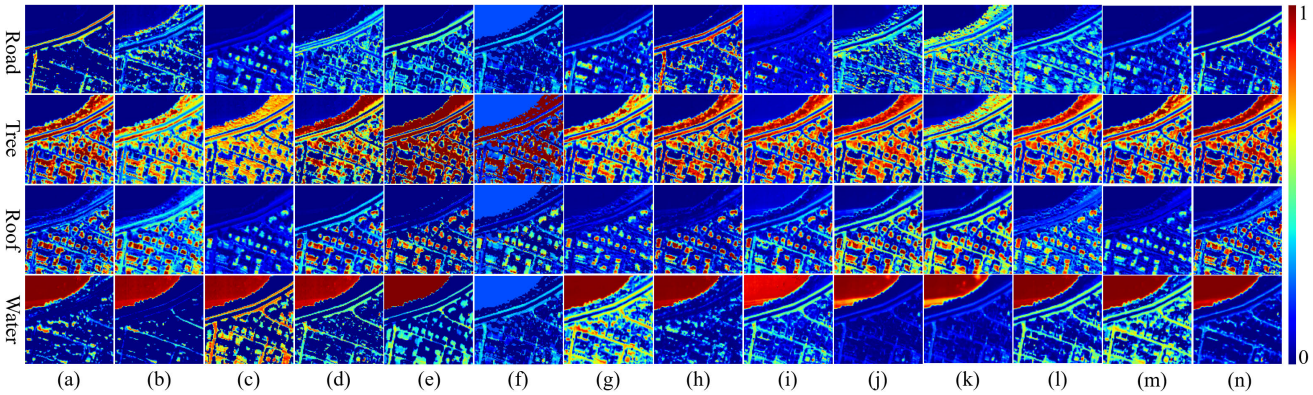


Fig. 10. Estimated abundances of different methods on the apex dataset. (a) Reference. (b) FCLSU. (c) SCLSU. (d) NUSAL. (e) DAEU. (f) SIDAEU. (g) MLAEM. (h) DeepTrans. (i) MLM-1DAE. (j) A2SN. (k) A2SAN. (l) SWCNet. (m) SRViT. (n) UNTR.

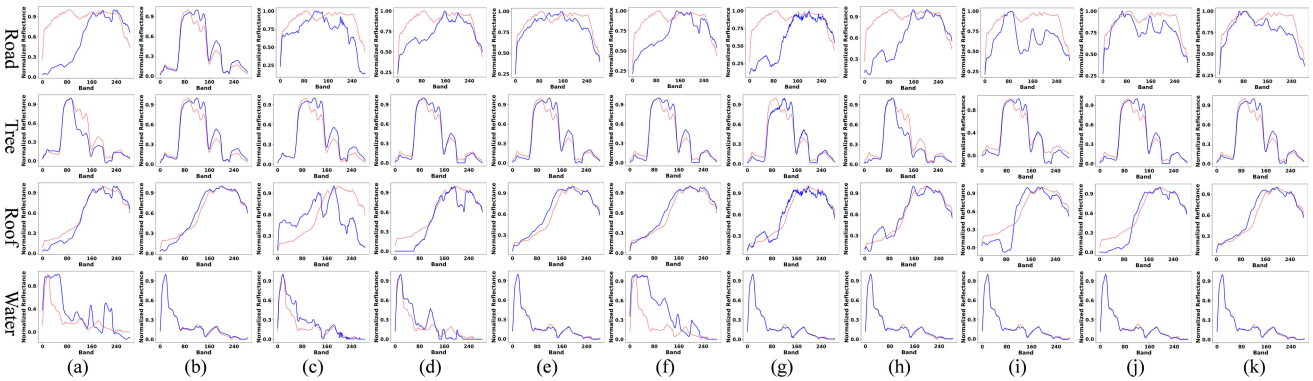


Fig. 11. Estimated endmembers of different methods on the apex dataset. (a) VCA. (b) DAEU. (c) SIDAEU. (d) MLAEM. (e) DeepTrans. (f) MLM-1DAE. (g) A2SN. (h) A2SAN. (i) SWCNet. (j) SRViT. (k) UNTR.

D. Real Data Results

We conduct comprehensive experiments on three real datasets using a wide range of comparison methods. The results for the Jasper Ridge dataset are shown in Table II, where the proposed UNTR method achieves the lowest aRMSE and aSAD of 0.0569 and 0.0350, respectively. The lowest abundance estimation errors are observed in each category, showing a significant reduction compared to other methods. For endmember estimation, the best results are achieved for the tree and road categories, while DAEU and DeepTrans lead in water and soil, respectively. Nevertheless, UNTR performs comparably, remaining close to these leading methods. In addition, the specifically estimated abundance maps and endmembers by different methods are shown in Figs. 7 and 8. Visually, the abundance maps from UNTR closely resemble the reference values and maintain consistency in local details. For endmembers, the estimated spectra from UNTR (blue curves) align well with the reference spectra (red curves), with notably smaller deviations in the road

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TABLE III

RMSE AND SAD RESULTS FOR THE APEX DATASET. MINIMUM VALUES FOR EACH ENDMEMBER ARE IN BOLD, WITH THE AVERAGE AS THE OVERALL METRIC. THE BEST RESULT IS SHOWN IN RED WITH ①, AND THE SECOND-BEST IN BLUE WITH ②

Methods	Initialization	RMSE					SAD				
		Road	Tree	Roof	Water	aRMSE	Road	Tree	Roof	Water	aSAD
FCLSU	VCA	0.2449	0.1324	0.1127	0.1314	0.1554	0.5467	0.2296	0.1138	0.4930	0.3458
SCLSU	VCA	0.2269	0.2236	0.2353	0.3545	0.2600	0.5467	0.2296	0.1138	0.4930	0.3458
NUSAL	VCA	0.2456	0.2568	0.1167	0.1977	0.2042	0.5467	0.2296	0.1138	0.4930	0.3458
DAEU	VCA	0.1642	0.1726	0.2069	0.2249	0.1922	0.1073	0.1446	0.1005	0.0499	0.1006
SIDAEU	VCA	0.1636	0.2017	0.2522	0.3869	0.2511	0.1974	0.1604	0.5148	0.2523	0.2812
MLAEM	VCA	0.1955	0.1358	0.2400	0.2846	0.2140	0.1578	**0.1006**	0.2130	0.2750	0.1866
DeepTrans	VCA	0.1929	0.1073	0.1279	**0.0864**	0.1286②	**0.0774**	0.1401	0.1084	0.0466	**0.0932**①
MLM-1DAE	VCA	0.2263	0.0982	0.1700	0.2086	0.1758	0.2214	0.1212	**0.0770**	0.4937	0.2283
A2SN	–	0.1886	0.1541	0.1340	0.1714	0.1620	0.4324	0.2068	0.1155	**0.0389**	0.1984
A2SAN	–	0.3081	0.2929	0.1371	0.1701	0.2270	0.4185	0.1906	0.1091	0.0455	0.1909
SWCNet	VCA	0.1938	0.1523	0.1524	0.2209	0.1798	0.1774	0.1918	0.2552	0.0416	0.1665
SRViT	VCA	0.1904	**0.0876**	0.2066	0.2300	0.1787	0.0893	0.1282	0.1959	0.0430	0.1141
UNTR	Random	**0.1517**	0.1155	**0.0930**	0.1160	**0.1190**①	0.0990	0.1335	0.1128	0.0402	0.0964②

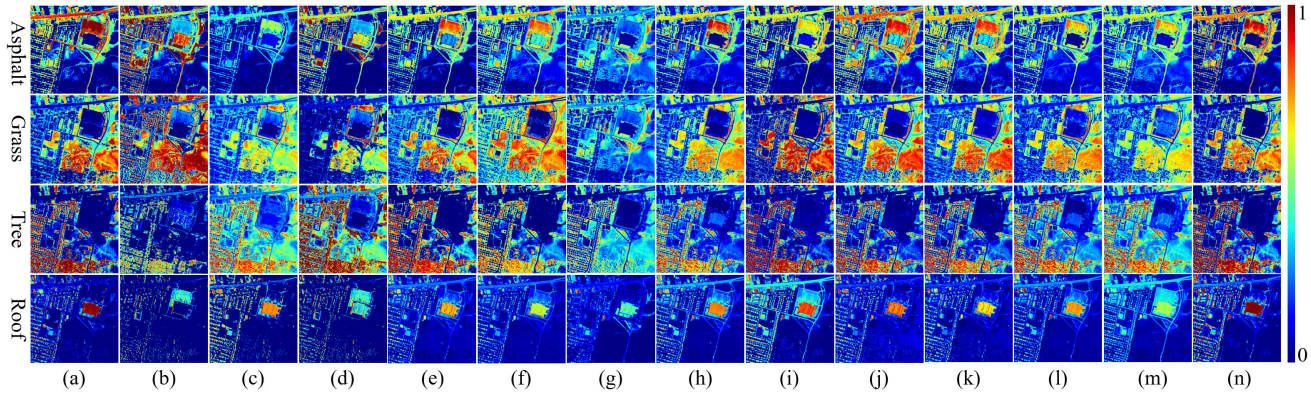


Fig. 12. Estimated abundances of different methods on the urban dataset. (a) Reference. (b) FCLSU. (c) SCLSU. (d) NUSAL. (e) DAEU. (f) SIDAEU. (g) MLAEM. (h) DeepTrans. (i) MLM-1DAE. (j) A2SN. (k) A2SAN. (l) SWCNet. (m) SRViT. (n) UNTR.

category, which tends to show larger errors in other methods. Furthermore, we perform principal component analysis (PCA) on all pixels, reference endmembers, and estimated endmembers from different methods on the Jasper Ridge dataset, and present the 2-D visualization in Fig. 9. The results show that the endmembers estimated by UNTR are consistently closest to the labeled ones, highlighting the effectiveness of its adaptive mixing mechanism modeling.

Table III shows the results for the Apex dataset, recently introduced for unmixing validation. Its complexity highlights the unique strengths of each method. Quantitative analysis shows that the proposed UNTR achieves the best performance in abundance estimation, with the lowest aRMSE of 0.1190. While DeepTrans excels in endmember estimation, the proposed UNTR attains an aSAD of 0.0964, exhibiting some variability primarily in the road category but outperforming DeepTrans in other categories such as road and water. Meanwhile, the proposed UNTR method achieves this unmixing performance without relying on well-initialized endmembers, significantly outperforming A2SN and A2SAN, which also do not depend on endmember initialization. This result underscores the feasibility of the endmember query mechanism and the effectiveness of the adaptive mixing mechanism modeling in UNTR. The detailed abundance maps and estimated spectra for each method are illustrated in Figs. 10 and 11.

Considering the comparison of various BHU methods on larger spatial extents and datasets with more texture information, Table IV presents the results for the Urban dataset. The proposed UNTR method achieves the lowest aRMSE of 0.0935 and aSAD of 0.0499. The aSAD is notably lower than that of other methods, with a substantial reduction compared to the suboptimal MLM-1DAE. As shown in Figs. 12 and 13, the estimated abundances by UNTR align much more closely with the labeled reference. Furthermore, the estimated endmembers are highly consistent with the true ones, providing a significantly better fit than other methods.

The proposed UNTR does not employ complex spatial–spectral feature extraction techniques for HSI, relying instead on convolutional operations. Nevertheless, the experimental results on three real datasets validate its effectiveness in achieving both high-precision endmember estimation and accurate abundance estimation without requiring well-initialized endmembers. This demonstrates that the focus on adaptive modeling of the mixing mechanism effectively reduces errors in endmember and abundance estimation caused by discrepancies between predefined mixing assumptions and the actual mixing mechanism, thereby enhancing unmixing performance. In addition, the endmember query mechanism mitigates perturbation effects

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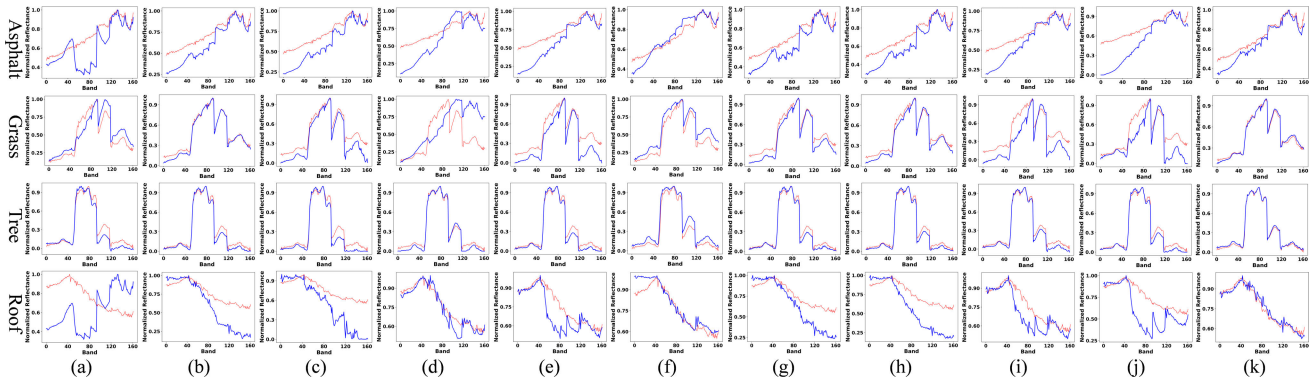


Fig. 13. Estimated endmembers of different methods on the urban dataset. (a) VCA. (b) DAEU. (c) SIDAEU. (d) MLAEM. (e) DeepTrans. (f) MLM-1DAE. (g) A2SN. (h) A2SAN. (i) SWCNet. (j) SRViT. (k) UNTR.

TABLE IV

RMSE AND SAD RESULTS FOR THE URBAN DATASET. MINIMUM VALUES FOR EACH ENDMEMBER ARE IN BOLD, WITH THE AVERAGE AS THE OVERALL METRIC. THE BEST RESULT IS SHOWN IN RED WITH ①, AND THE SECOND-BEST IN BLUE WITH ②

Methods	Initialization	RMSE					SAD				
		Asphalt	Grass	Tree	Roof	aRMSE	Asphalt	Grass	Tree	Roof	aSAD
FCLSU	VCA	0.2891	0.2880	0.2764	0.2205	0.2685	0.2028	0.1990	0.1744	0.4672	0.2608
SCLSU	VCA	0.2222	0.1527	0.2139	0.0854	0.1685	0.2028	0.1990	0.1744	0.4672	0.2608
NUSAL	VCA	0.2664	0.2331	0.2638	0.1583	0.2304	0.2028	0.1990	0.1744	0.4672	0.2608
DAEU	VCA	0.1182	**0.1040**	0.0908	0.1258	0.1097	0.2020	**0.0369**	0.0889	0.3340	0.1654
SIDAEU	VCA	0.1272	0.1464	0.1196	0.1097	0.1257	0.1741	0.2153	0.1698	0.4164	0.2439
MLAEM	VCA	0.2837	0.3161	0.2527	0.1727	0.2563	0.2671	0.4478	0.0987	0.0592	0.2182
DeepTrans	VCA	0.1530	0.1192	0.1299	0.1043	0.1266	0.2247	0.2692	0.1175	0.1261	0.1844
MLM-1DAE	VCA	0.1384	0.1394	0.0839	0.1326	0.1236	0.0832	0.1039	0.1264	0.0510	0.0912②
A2SN	–	0.1219	0.1308	0.1055	**0.0707**	0.1073	0.1069	0.1395	0.0991	0.2290	0.1436
A2SAN	–	0.1137	0.1219	0.0961	0.0822	0.1035②	0.1195	0.0957	0.0864	0.3156	0.1543
SWCNet	VCA	0.1428	0.1507	0.1335	0.1038	0.1327	0.1860	0.3349	0.1075	0.0993	0.1819
SRViT	VCA	0.2031	0.1925	0.1719	0.1545	0.1805	0.3663	0.3246	0.1110	0.2590	0.2652
UNTR	Random	**0.1050**	0.1095	**0.0674**	0.0922	**0.0935①**	**0.0789**	0.0451	**0.0419**	**0.0337**	**0.0499①**

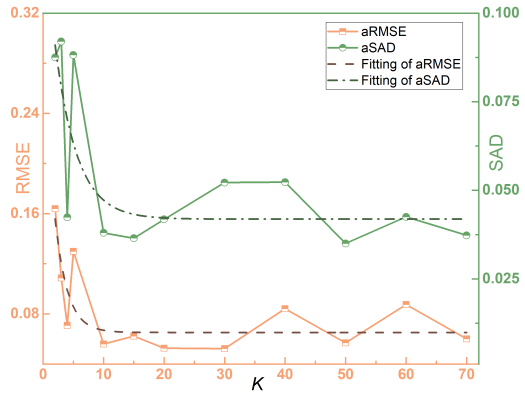


Fig. 14. Impact of K on the jasper ridge dataset.

and optimizes random initialization data toward the image data distribution, significantly alleviating the reliance on well-initialized endmembers.

IV. DISCUSSION

A. Impact of K in the Endmember Query Mechanism

The parameter K is critical for both the endmember query mechanism and endmember estimation. From a statistical perspective, representing each endmember with a single sequence leads to high estimation variance, whereas averaging multiple sequences effectively reduces this variance and brings the expected value closer to the reference endmember. Theoretically, increasing K should substantially reduce estimation error at first, followed by gradual stabilization. To validate this, we conduct experiments on the Jasper Ridge dataset with varying K values, as shown in Fig. 14.

It can be observed that for single-digit values of K, perturbations remain substantial, though a clear trend of error reduction is evident. When K reaches 20, the errors of endmembers and their corresponding abundances begin to stabilize. To highlight this trend, curve fitting is applied to both the aRMSE and aSAD metrics. These results align with the theoretical expectation of reduced estimation variance, thereby validating the effectiveness of the proposed approach.

B. Ablation Study

To validate the effectiveness of each module, an ablation study is conducted, as shown in Table V. In the table, "AMMMM" denotes the adaptive mixing mechanism modeling module, and "EQM" represents the endmember query mechanism. When neither module is used, the model is constructed based solely on LMM and does not incorporate the proposed distribution alignment constraint.

The results indicate that the unmixing model relying solely on randomly initialized endmembers shows limited

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TABLE V
RESULTS OF ABLATION STUDY FOR EACH MODULE OF UNTR

Datasets	AMMMM	EQM	aRMSE	aSAD
Jasper Ridge	✗	✗	0.1468	0.2538
	✗	✓	0.0609	0.0707
	✓	✗	**0.0531**	0.0612
	✓	✓	0.0569	**0.0350**
Apex	✗	✗	0.2357	0.2120
	✗	✓	0.1392	0.1061
	✓	✗	0.1208	0.1556
	✓	✓	**0.1190**	**0.0964**
Urban	✗	✗	0.1560	0.2381
	✗	✓	0.1051	0.0519
	✓	✗	0.1113	0.0720
	✓	✓	**0.0935**	**0.0499**

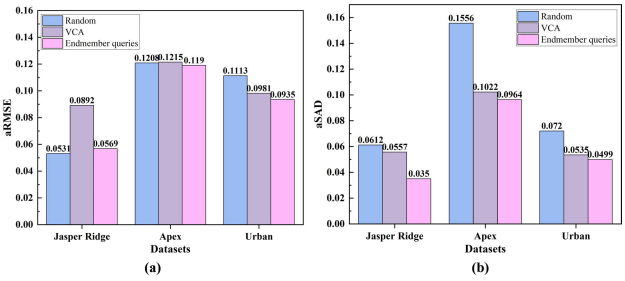


Fig. 15. Comparison of different initialization strategies. (a) aRMSE. (b) aSAD.

performance under the LMM assumption on the three real datasets. In contrast, incorporating the endmember query mechanism significantly enhances unmixing performance, as it allows the randomly initialized endmembers to be adaptively optimized toward the image data distribution. Furthermore, unmixing performance significantly improves with the integration of the adaptive mixing mechanism modeling module.

C. Comparison of Different Initialization Strategies

A comparison of different initialization strategies is presented in Fig. 15. Here, "Random" and "VCA" refer to the cases where each endmember is represented as a single trainable random sequence under random initialization and VCA initialization, respectively. "Endmembers queries" represents the endmember query mechanism.

Under the distribution alignment constraint, random initialization of endmembers achieves effective convergence within the adaptive mixing mechanism modeling framework, yielding well-estimated endmembers and abundances at a leading performance level. This validates the feasibility of the proposed mixing mechanism modeling. In addition, the VCA initialization employs the common strategy of freezing endmembers during initial optimization, followed by co-training with other parameters after unfreezing. While this approach enhances unmixing performance, the improvement is modest, demonstrating that the model can still converge effectively without well-initialized endmembers. The proposed endmember queries, which represent a single endmember using multiple trainable random sequences, capture the diverse

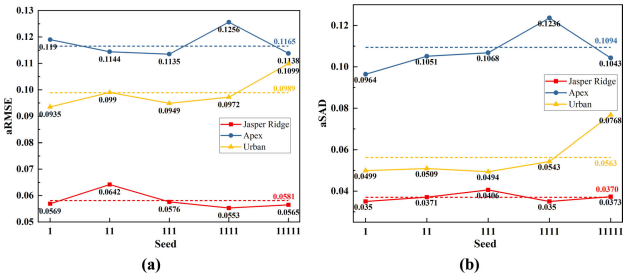


Fig. 16. Comparison of different random seeds. (a) aRMSE. (b) aSAD.

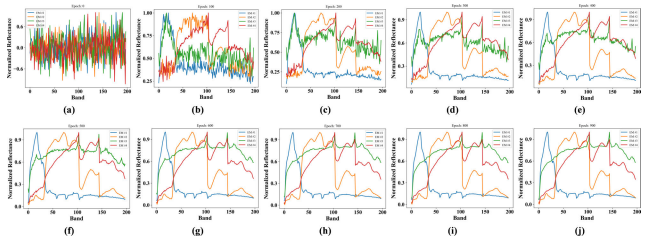


Fig. 17. Endmember optimization visualization on the Jasper Ridge dataset. (a) Random initialization. (b) Epoch: 100. (c) Epoch: 200. (d) Epoch: 300. (e) Epoch: 400. (f) Epoch: 500. (g) Epoch: 600. (h) Epoch: 700. (i) Epoch: 800. (j) Epoch: 900.

endmember features synthesized into the endmembers without relying on initialization.

D. Stability of Experimental Results

To further validate the unmixing performance of UNTR under different random initialization, we employ varying random seeds, as illustrated in Fig. 16, where the dashed line represents the average results obtained from these seeds. The results demonstrate that, despite the influence of different random seeds, the unmixing outcomes across the three real datasets remain stable and consistently converge. This stability highlights the effectiveness of the constructed model, which successfully optimizes the random sequences toward the image data distribution under the distribution alignment constraint. Moreover, the mutual constraints and facilitation between endmember estimation and abundance estimation in BHU ensure robustness, reflecting the overall stability of the unmixing process under random initialization.

E. Endmember Visualization During Training

We visualized the endmember optimization process during the training on the Jasper Ridge dataset, as depicted in Fig. 17. It illustrates how the endmember estimation evolves throughout the training, demonstrating the convergence and refinement of the endmember spectra as the model progresses. It can be observed that the endmember queries are randomly initialized, resulting in initially random endmember estimation. However, as training progresses, the endmember queries are gradually optimized, leading to endmembers that become increasingly distinguishable and eventually converge toward the real endmembers. By the 100th iteration, the endmembers are already differentiated, and with further iterations, they become visually distinct and stabilize. This demonstrates the effectiveness of UNTR for endmember estimation without

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TABLE VI

PARAMETERS AND FLOPS ON THE JASPER RIDGE DATASET

Metircs	DAEU	DeepTrans	MLM-1DAE	A2SN	A2SAN	SWCNet	SRViT	UNTR
Params	0.009M	7.542M	0.116M	0.044M	0.043M	1.126M	63.802M	5.929M
FLOPs	18.33K	2.86G	0.23M	0.91G	0.90G	22.51G	400.87G	36.94G

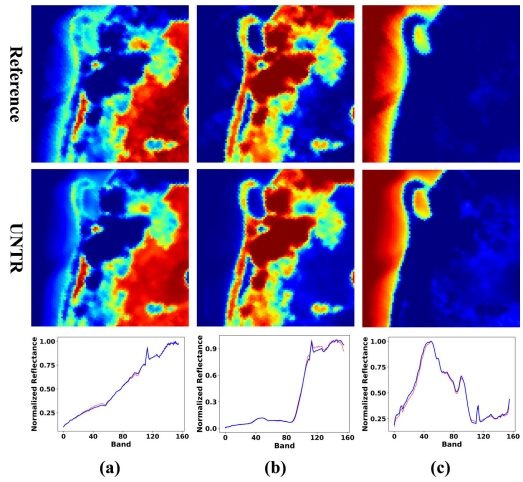


Fig. 18. Estimated abundances and endmembers of UNTR on the Samson dataset. (a) Soil. (b) Tree. (c) Water.

relying on any initialization method, highlighting its strong BHU capabilities.

F. Analysis of Model Complexity

We further analyze the number of parameters and floating point operations (FLOPs) of different methods using the Jasper Ridge dataset as an example, as shown in Table VI. Compared to pixel-level methods, image-level unmixing methods exhibit an increase in both parameters and FLOPs, but deliver superior unmixing performance. UNTR reduces the number of parameters while increasing FLOPs compared to Transformer-based DeepTrans, and achieves reductions in both compared to SRViT, which is also built upon the Transformer architecture. The reduction in parameters of UNTR is attributed to the use of straightforward convolution for spatial–spectral feature extraction without complex designs. However, the increase in FLOPs stems from its approach of estimating the inherent mixing mechanism of HSI using an attention mechanism–a global structure that calculates the correlation between global abundances and endmember queries. Considering the observed improvement in unmixing performance, the feasibility of mixing mechanism modeling is validated, though further research is needed to explore FLOPs reduction.

G. Experiments on Other Datasets

We provide experimental results on additional real datasets to further demonstrate the effectiveness of the proposed UNTR method, including Samson with three endmembers, as well as Urban and Washington DC Mall, each labeled with six endmembers. Detailed abundance and endmember quantitative evaluation metrics on the Samson and Urban6 datasets are given in Table VII. Furthermore, Figs. 18 and 19 present the visualization results for abundances and endmembers on

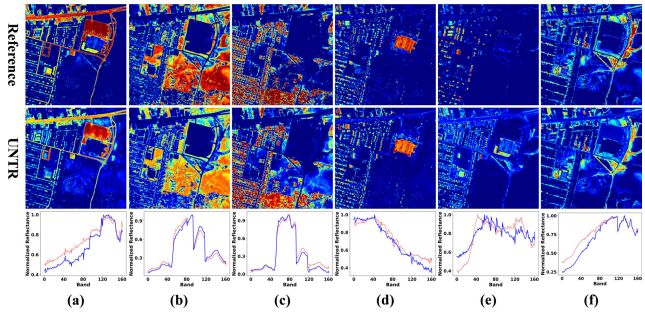


Fig. 19. Estimated abundances and endmembers of UNTR on the urban6 dataset. (a) Asphalt. (b) Grass. (c) Tree. (d) Roof. (e) Metal. (f) Dirt.

TABLE VII

RESULTS OF UNTR FOR THE SAMSON AND URBAN6 DATASETS

Datasets	Endmembers	RMSE	SAD
Samson	Soil	0.0686	0.0156
	Tree	0.0338	0.0329
	Water	0.0558	0.0279
	Average	**0.0527**	**0.0255**
Urban6	Asphalt	0.1368	0.0837
	Grass	0.1422	0.0947
	Tree	0.1052	0.1113
	Roof	0.0608	0.0749
	Metal	0.1461	0.1119
	Dirt	0.0755	0.1100
	Average	**0.1111**	**0.0978**

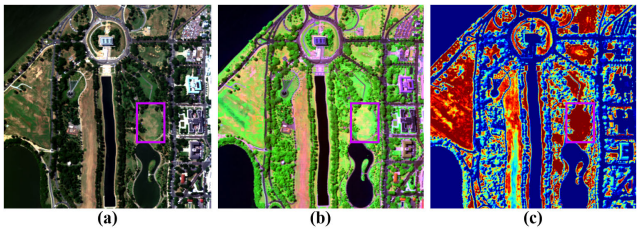


Fig. 20. Differences between labeled grass and HSI on the Washington DC mall dataset. (a) True-color image. (b) False-color image. (c) Labeled grass.

the Samson and Urban6 datasets, respectively, showcasing the performance of the proposed UNTR method without relying on any well-initialized endmembers.

We also conduct experiments on the Washington DC Mall dataset. According to the description in [16], the endmembers of this dataset were collected from the image, while the abundances were estimated using FCLSU. Visualization of the original HSI with the labeled grass is depicted in Fig. 20. As shown in the purple box, there is a notable regional inconsistency between the true and false-color images of the hyperspectral data and the ground truth, particularly in areas labeled as grass. Therefore, the unmixing results for this dataset were evaluated solely through visual inspection.

Fig. 21 shows the visualized abundances and endmembers estimated by the proposed method on the Washington DC Mall dataset. Compared to the abundances estimated by FCLSU, UNTR classifies grass into two distinct categories and merges the roof and road into one category. Intuitively, the results for the two grass categories exhibit regional consistency with

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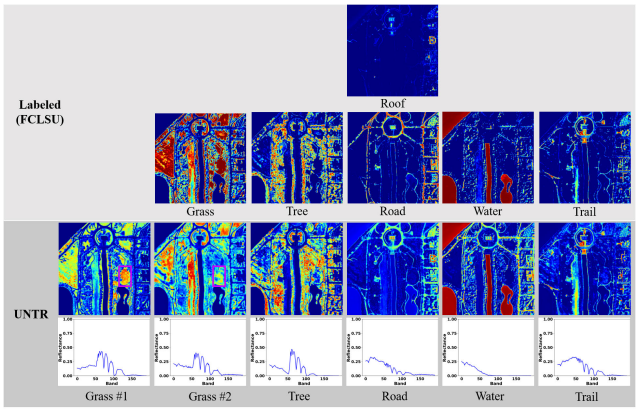


Fig. 21. Estimated abundances and endmembers of UNTR on the washington DC Mall dataset.

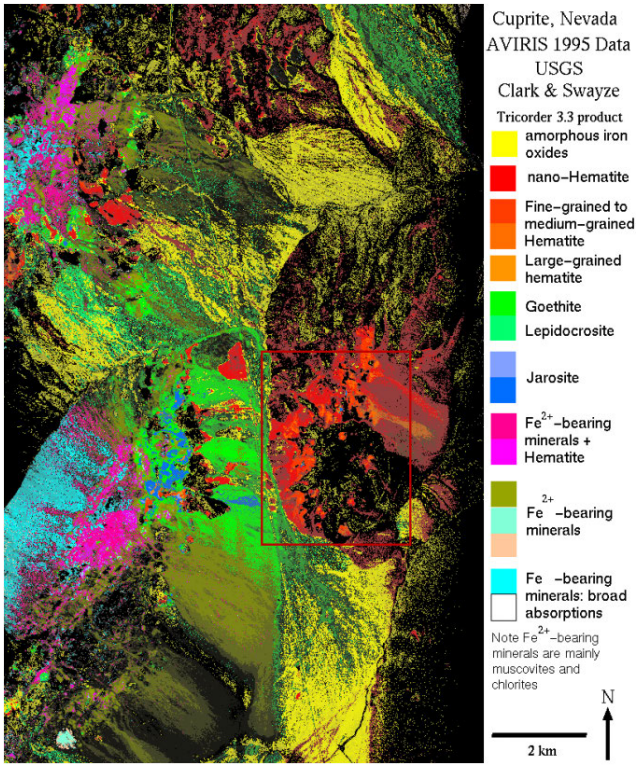


Fig. 22. USGS mineral map of the cuprite mining district.

the original HSI in Fig. 20. While the image contains fewer roof areas, UNTR tends to merge them with the road category due to their similarities. This illustrates the characteristic of UNTR, where the simultaneous estimation of endmembers and their abundances from the image—without any prior information—results in the dominance of widely varying endmembers, while maintaining separability for less differentiated endmembers, such as grass #1 and grass #2.

H. Result on Cuprite Dataset

We further evaluate the proposed unmixing method on the widely used Cuprite dataset. A commonly used subimage is adopted for benchmarking, consisting of 250×191 pixels with 188 spectral bands retained for analysis. The mineral distribution map for this district was generated in 1995, whereas the HSI was acquired in 1997; therefore, Fig. 22 is provided for qualitative visualization only [64].

In our experiments, the number of endmembers is set to 12 based on HySime [65], and prior studies [64]. Following previous work [44], six representative materials (Alunite, Sphene, Kaolinite1, Sphene2, Montmorillonite, and Kaolinite2) are visualized in Fig. 23. Compared with the reference maps, the abundances produced by UNTR are generally consistent for these representative materials and preserve detailed spatial structures. In particular, the abundance maps for all 12 endmember categories estimated by UNTR are shown in Fig. 24(a). The corresponding classification map, obtained by assigning each pixel to the category with the maximum abundance, is presented in Fig. 24(b). Fig. 24(c) presents the abundance stack map, generated by overlaying the 12 abundance maps with transparency determined by their respective abundance values. The resulting visualization shows a spatial distribution similar to that of the reference image in Fig. 22.

I. Mismatched Endmember Numbers

The number of endmembers has a noticeable impact on unmixing performance. Nevertheless, UNTR maintains stable and effective unmixing results on the Urban dataset under varying numbers of endmembers. To further investigate the influence of mismatched endmember numbers, we conduct experiments within a controlled range. Using the Jasper Ridge dataset as an example, the number of endmembers R is varied from 3 to 7. The corresponding abundance estimation results are presented in Fig. 25.

An interesting behavior can be observed as the number of endmembers varies. When $R = 3$, UNTR tends to merge road and soil regions. At $R = 5$, the boundary between water and soil becomes distinguishable, while other categories remain largely unchanged. When $R = 6$, additional road structures are further separated compared with the $R = 5$ case. At $R = 7$, the boundary between water and tree regions is further delineated. Similar trends are observed on the Apex dataset. These results indicate that, within a certain range, increasing the mismatched number of endmembers in UNTR leads to more refined unmixing results. While the primary material categories are preserved, the model is able to progressively reveal finer substructures within each category.

J. Discussion on Distribution Alignment Constraint

In this article, the minimum volume constraint $\mathcal{L}_{\mathrm{MV}}$ [47] is intuitively adopted as the distribution alignment constraint $\mathcal{L}_{\mathrm{DA}}$, primarily owing to its effectiveness and consistency with traditional unmixing methods. An intuitive interpretation is that the minimum volume constraint enforces endmember consistency, driving different endmembers to converge toward the center of the data distribution. A natural extension of this optimization is to introduce endmember diversity, which preserves a certain degree of variation among endmembers. This enables them to approach the data center from different directions, thereby effectively forming a minimum-volume simplex. Let the estimated endmembers be denoted by $\hat{\mathbf{E}}$. The

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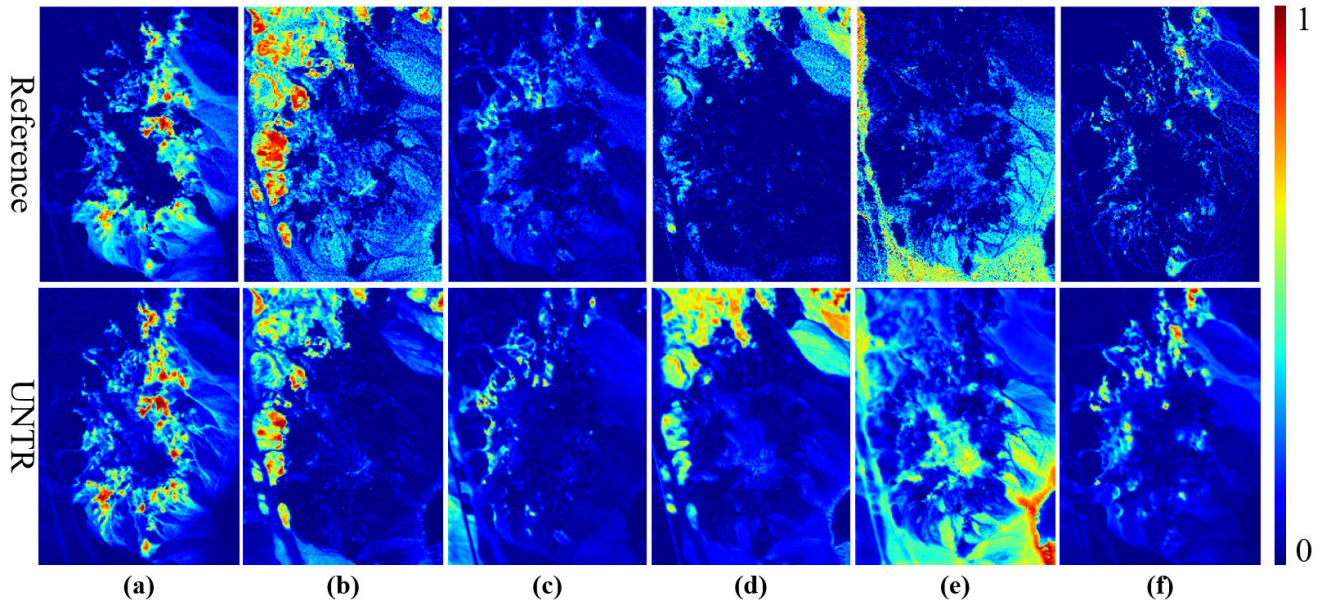


Fig. 23. Estimated abundances for representative materials on the cuprite dataset. (a) Alunite. (b) Sphene. (c) Kaolinite1. (d) Sphene2. (e) Montmorillomnite. (f) Kaolinite2.

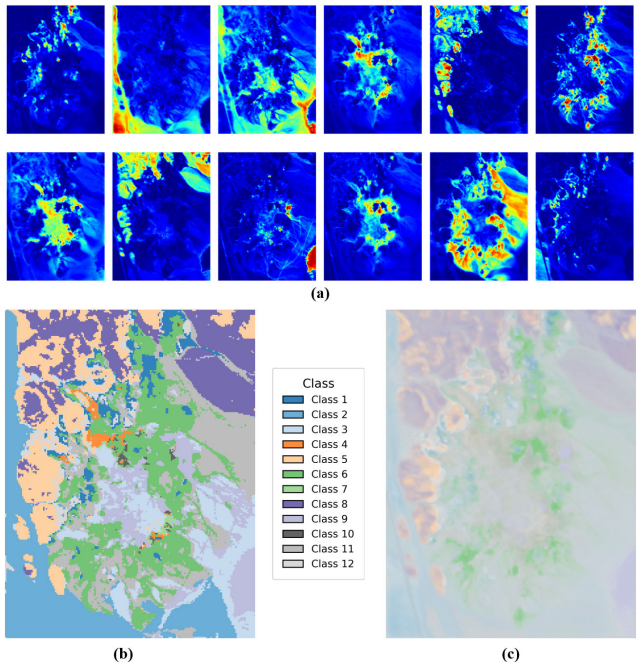


Fig. 24. Visualization results of UNTR on the cuprite dataset. (a) Estimated abundances for 12 materials on the cuprite dataset. (b) Classification map. (c) Abundance stack map.

endmember diversity constraint, $\mathcal{L}_{\mathrm{ED}}$, can then be formulated as

$$\mathcal{L}_{\mathrm{ED}} = \frac{1}{R(R-1)} \sum_{i=1}^{R} \sum_{\substack{j=1 \\ j \neq i}}^{R} \left| \frac{\hat{\mathbf{E}}_i \cdot \hat{\mathbf{E}}_j}{\left\| \hat{\mathbf{E}}_i \right\|_2 \left\| \hat{\mathbf{E}}_j \right\|_2} \right| \tag{19}$$

where R denotes the number of endmembers. By minimizing the cosine similarity, the distances between endmembers are increased, thereby encouraging greater differentiation among them.

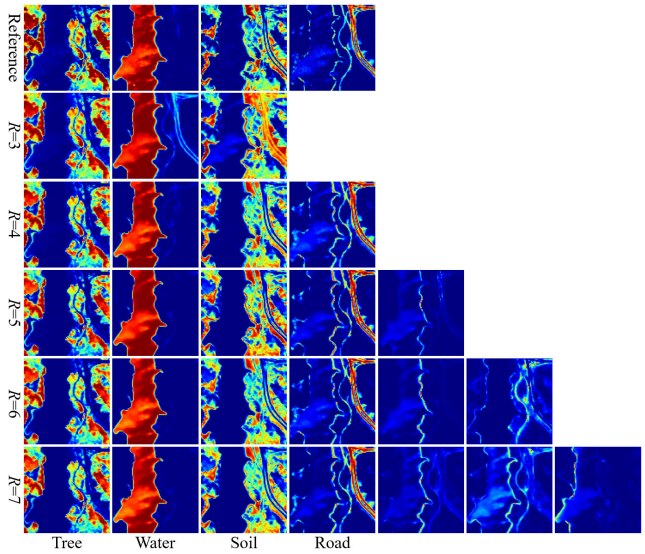


Fig. 25. Abundance maps of UNTR with different numbers of endmembers.

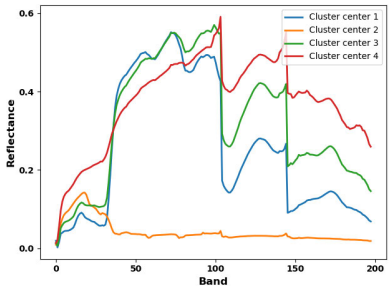


Fig. 26. Cluster centers on the Jasper Ridge dataset.

The aforementioned constraints are all defined at the level of individual endmembers. To further align the endmember distribution with the image distribution, a moment-matching constraint is introduced. This constraint achieves distributional alignment by minimizing the discrepancies in the first N-order

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TABLE VIII
RESULTS FOR JASPER RIDGE DATASET UNDER DIFFERENT CONSTRAINTS

$\mathcal{L}_{MV}$	$\mathcal{L}_{MM}$	$\mathcal{L}_{ED}$	aRMSE	aSAD
✓	✗	✗	0.0795 ± 0.0254	0.0559 ± 0.0244
✓	✗	✓	0.0611 ± 0.0082	0.0557 ± 0.0151
✗	✓	✗	0.0857 ± 0.0058	0.0966 ± 0.0124
✗	✓	✓	0.0919 ± 0.0091	0.1206 ± 0.0147
✓	✓	✗	0.0750 ± 0.0251	0.0557 ± 0.0218
✓	✓	✓	0.0786 ± 0.0288	0.0540 ± 0.0197
✓	✓(cluster)	✓	**0.0608 ± 0.0077**	**0.0447 ± 0.0068**

statistical moments between the endmembers and the image, as formulated below

$$\mathcal{L}_{MM} = \frac{1}{N}\sum_{n=1}^{N}\left\|\mathbf{m}^{(n)}\left(\hat{\mathbf{E}}\right) - \mathbf{m}^{(n)}\left(\mathbf{I}\right)\right\|_2^2 \tag{20}$$

where $\mathbf{m}^{(n)}$ denotes the bandwise statistical moment vector. Furthermore, the moment-matching constraint establishes a form of alignment between the endmember distribution and the image data distribution; however, despite this alignment, the two distributions still differ in both scale and statistical characteristics. Considering that clustering is an unsupervised classification process, each cluster center can, to a certain extent, serve as a proxy for an endmember, as shown in Fig. 26. For the moment-matching constraint, the statistical moments of the cluster centers can also be used as a proxy for the image data distribution.

Here, taking the Jasper Ridge dataset as an example, we conduct five Monte Carlo experiments with random seeds ranging from 1 to 5. The corresponding statistical results are reported in Table VIII. By incorporating $\mathcal{L}_{MM}$ and $\mathcal{L}_{ED}$ into the distribution alignment constraint, the overall performance is further improved.

V. CONCLUSION

In this article, we propose UNTR, a novel BHU method that represents the first attempt to model the mixing mechanism directly from the image itself. Experimental results on several synthetic and real datasets demonstrate that UNTR achieves state-of-the-art unmixing performance without relying on predefined mixing assumptions or well-initialized endmembers. This validates the feasibility of the proposed adaptive mixing mechanism modeling approach, which mitigates biases in endmember and abundance estimation caused by discrepancies between the predefined assumptions and the actual mixing mechanism. Furthermore, the proposed endmember query mechanism effectively optimizes random initialization sequences toward the HSI data distribution, significantly alleviating the dependence on well-initialized endmembers. Without any prior information, UNTR achieves leading performance and offers a new perspective for BHU research.

APPENDIX A
VISUALIZATION RESULTS ON SYNTHETIC DATASETS

The false-color images and the endmembers used for data synthesis are illustrated in Fig. 5. For the different datasets, abundance distributions are generated at varying levels to comprehensively evaluate the performance of the proposed method. The statistical details of the synthetic datasets are summarized in Table IX.

TABLE IX
STATISTICAL DETAILS OF THE SYNTHETIC DATASETS

Datasets	Pure pixels	High abundance pixels (>0.9)
LMM	0.08%	5.75%
Fan model	0.08%	8.37%
MLM	0.09%	4.57%

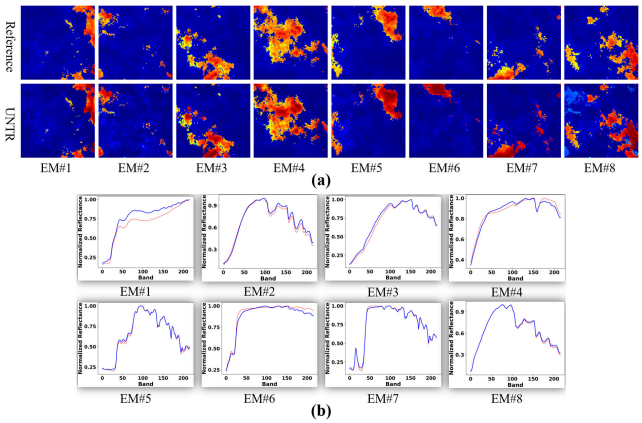


Fig. 27. Estimated (a) abundances and (b) endmembers of UNTR on the LMM-based synthetic dataset.

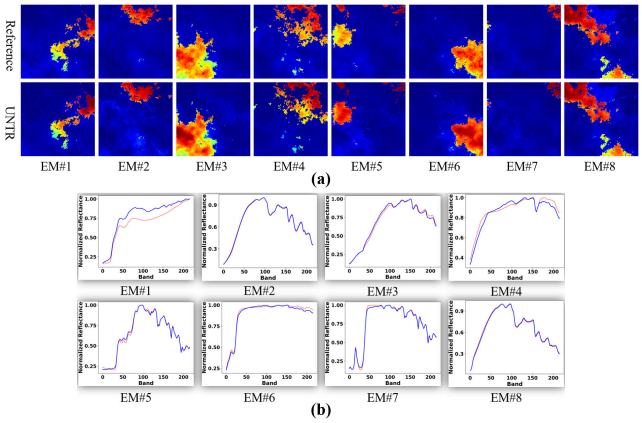


Fig. 28. Estimated (a) abundances and (b) endmembers of UNTR on the Fan model-based synthetic dataset.

Figs. 27–29 illustrate the abundance and endmember estimation results of the proposed method on the three synthetic datasets, together with comparisons to the corresponding ground truth.

APPENDIX B
RESULTS ON RAY-TRACING DATASET

Recently, the ray-tracing dataset generated using physically based ray tracer (PBRT) models has been increasingly adopted for validating nonlinear unmixing methods [66]. To further evaluate the effectiveness of the proposed approach, we conduct comparative experiments on this dataset. It consists of six scenes, as illustrated in Fig. 30, covering two material categories, namely beech and poplar. Excluding the homogeneous

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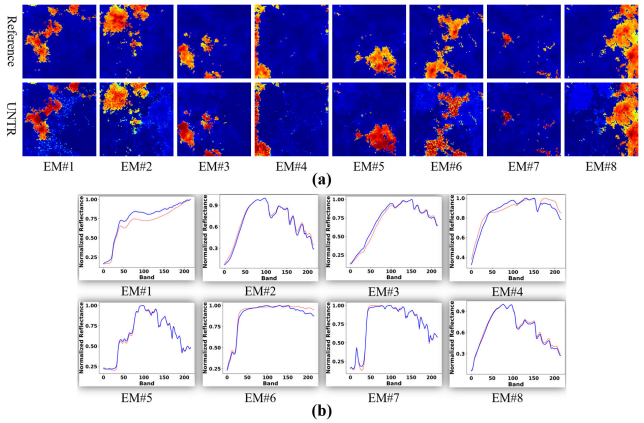


Fig. 29. Estimated (a) abundances and (b) endmembers of UNTR on the MLM-based synthetic dataset.

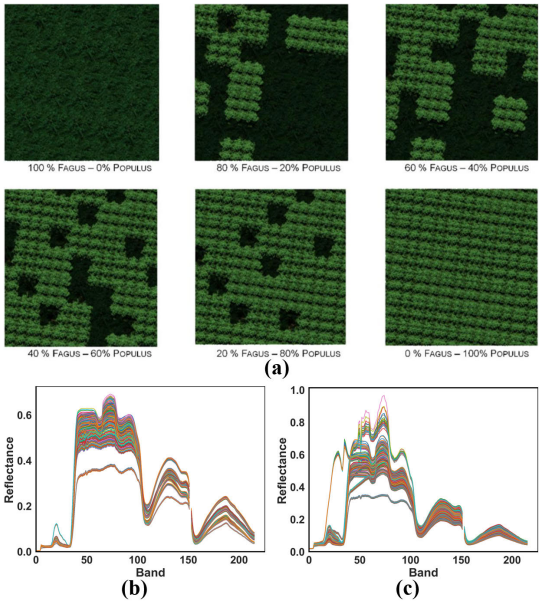


Fig. 30. Ray-tracing dataset. (a) High-resolution image with different proportions of beech and poplar trees. (b) Beech spectra for each pixel. (c) Poplar spectra for each pixel.

beech and poplar scenes, the remaining four scenes exhibit varying proportions of mixed materials.

This dataset exhibits significant spectral variability. Therefore, following [26], [44], RMSE and reconstruction error (RE) are adopted as evaluation metrics for unmixing performance. Each scene in the dataset contains 12×36 pixels, and after removing NaN values, 215 spectral bands are retained for the experiment. Due to the weak spatial correlation in this dataset [26], methods that rely heavily on spatial feature extraction, including the proposed approach and other image-processing-based DL methods, are not directly applicable. Accordingly, we design an alternative UNTR-1D that preserves the core concept of adaptive mixing mechanism modeling in UNTR. In particular, individual pixels are used as network inputs, and the encoder adopts the DAEU architecture [10]. The decoder concatenates 1-D abundance and endmember representations, followed by multiple fully connected layers, which jointly constitute the 1-D adaptive mixing mechanism modeling module. The quantitative results on this dataset are shown in Table X.

TABLE X

RMSE AND RE RESULTS FOR THE RAY-TRACING DATASET. THE BEST RESULT IS SHOWN IN RED, AND THE SECOND-BEST IN BLUE

Forest_b20p80						
Metrics	DAEU	MLAEM	MLM-1DAE	DAAN	DEAE	UNTR-1D
RMSE	0.3089	0.2208	0.2957	0.2494	0.0350	**0.0202**
RE	0.0873	0.0370	**0.0215**	0.0657	0.0586	0.0486
Forest_b40p60						
Metrics	DAEU	MLAEM	MLM-1DAE	DAAN	DEAE	UNTR-1D
RMSE	0.2243	0.2091	0.2247	0.1682	0.0531	**0.0392**
RE	0.0862	0.0390	**0.0185**	0.0565	0.0516	0.0842
Forest_b60p40						
Metrics	DAEU	MLAEM	MLM-1DAE	DAAN	DEAE	UNTR-1D
RMSE	0.2005	0.3033	0.2218	0.1481	**0.0563**	0.0816
RE	0.0512	0.0329	**0.0145**	0.0507	0.0437	0.0347
Forest_b80p20						
Metrics	DAEU	MLAEM	MLM-1DAE	DAAN	DEAE	UNTR-1D
RMSE	0.2845	0.2294	0.2829	0.0925	**0.0159**	0.0597
RE	0.0561	0.0288	**0.0115**	0.0428	0.0411	0.0310

As observed, UNTR-1D, which preserves the core concept of adaptive mixing mechanism modeling, demonstrates strong unmixing capability on this dataset. It achieves the best performance in the Forest_b20p80 and Forest_b40p60 scenes, outperforming all competing methods. In the Forest_b60p40 and Forest_b80p20 scenes, UNTR-1D ranks second, slightly behind DEAE. Regarding RE, UNTR-1D does not achieve the best performance across all four scenarios, as methods such as MLAEM and MLM-1DAE directly optimize RMSE as their training objective, inherently favoring lower REs. When considering methods that simultaneously incorporate SAD as a training constraint, UNTR-1D attains the lowest RE in two scenarios, while DEAE achieves the lowest RE in the remaining two.

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